Where does Qwen3-32B hold a paper's authors?

Stages 1–9

Small samples: find a signal, try to break it

20 → 50 → 200 papers

Stages 10–11

Full cohort: locate it, then intervene

9,108 papers · 149,999 name tokens

One question, four lines of argument

Each line narrows the space of explanations; small samples set it up, the full cohort settles it

dark pages = Stages 1–9 | light pages = Stages 10–11 | purple tag = comparison

Where does Qwen3-32B hold a paper's authors, and can it say them?

A · This paper?

Paper-specific, not an easy name?

- 1–9 · Stage 5, 7.2, 7.3, 8
- 10–11 · title signal, name prior, probe

B · Readout tool

What can each readout see?

- 1–9 · Stage 3, 7.4, 8, 9
- 10–11 · lens comparison, linear probe

C · Which layer

When does it appear, when is it used?

- 1–9 · Stage 4, 7.2, 7.3, 8 (L56)
- 10–11 · fine layers, patching, knockout

D · Which token

Which part of name or prompt carries it?

- 1–9 · Stage 6, 7.1
- 10–11 · name format, tail tokens, co-authors

Stages 1–9 · 20–200 papers

Pilot design

- Pilot50 / Stage 8: 50 and 200 papers
- Mostly first-listed author, first token
- Score: gold vs 1 or 100 false names (logit margin, win rate)
- A/B both R0, matched in pairs
- Wrong titles: 1 crossed or 20 shuffled
- Layer L56 chosen after repeated looks

Stages 10–11 · 9,108 papers

Full-cohort design

- All 9,108 papers, every author, every token
- Score: rank over the whole vocabulary, log10(1/rank)
- target = A + C, control = B + D (R0 + R1), compared within citation bins
- 5 matched false titles + no title
- All layers reported, fine layers added
- Parity and Stage 6 cross-check before analysis

What changed in the design, and why

Small samples could not separate "small effect" from "no effect" in a given layer. The full cohort trades candidate-based scores for full-vocabulary ranks and pays for it with GPU time. **Note:** target/control include R1 papers; with R0 only (A vs B, as in the pilot) the output gap is +0.199 vs +0.207 and every layer looks the same.

Same papers, two ways of grouping

Stages 1–9 compare A vs B; Stages 10–11 compare target vs control. Counts from the 6,841-paper cohort.

	recognised (target)	not recognised (control)
R0 · no true author written	A = 4,110	B = 2,186
R1 · some authors written (HR2 > 0.5)	C = 511	D = 34
Total	**target = A + C = 4,621**	**control = B + D = 2,220**

Stages 1–9

A vs B: both R0, so the contrast is purely "knows but cannot say"

Stages 10–11

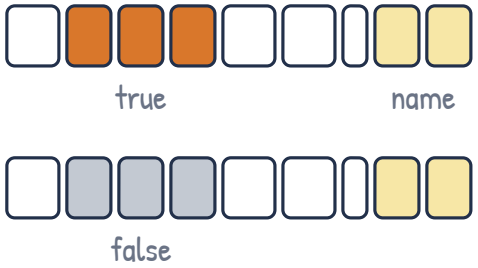
(A + C) vs (B + D): adds 511 partial-recall papers, almost all on the target side

Check done: with R0 only (A vs B) the output gap is +0.199, versus +0.207 for target vs control, and every layer looks the same. The two phases compare like with like.

Reading the method figures

1 2 3 steps, in order

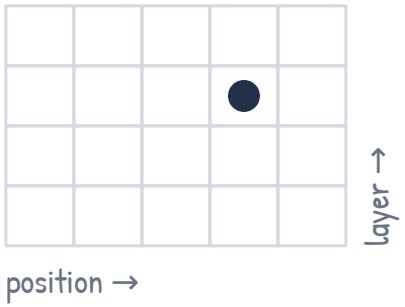
Prompt tokens



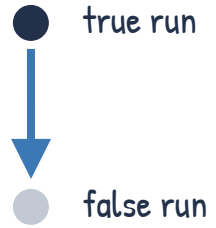
Layers



State = dot



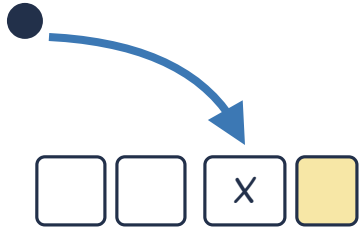
Copy a state



Block



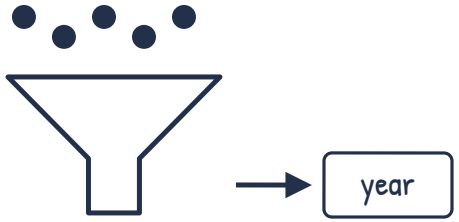
Inject



Read out



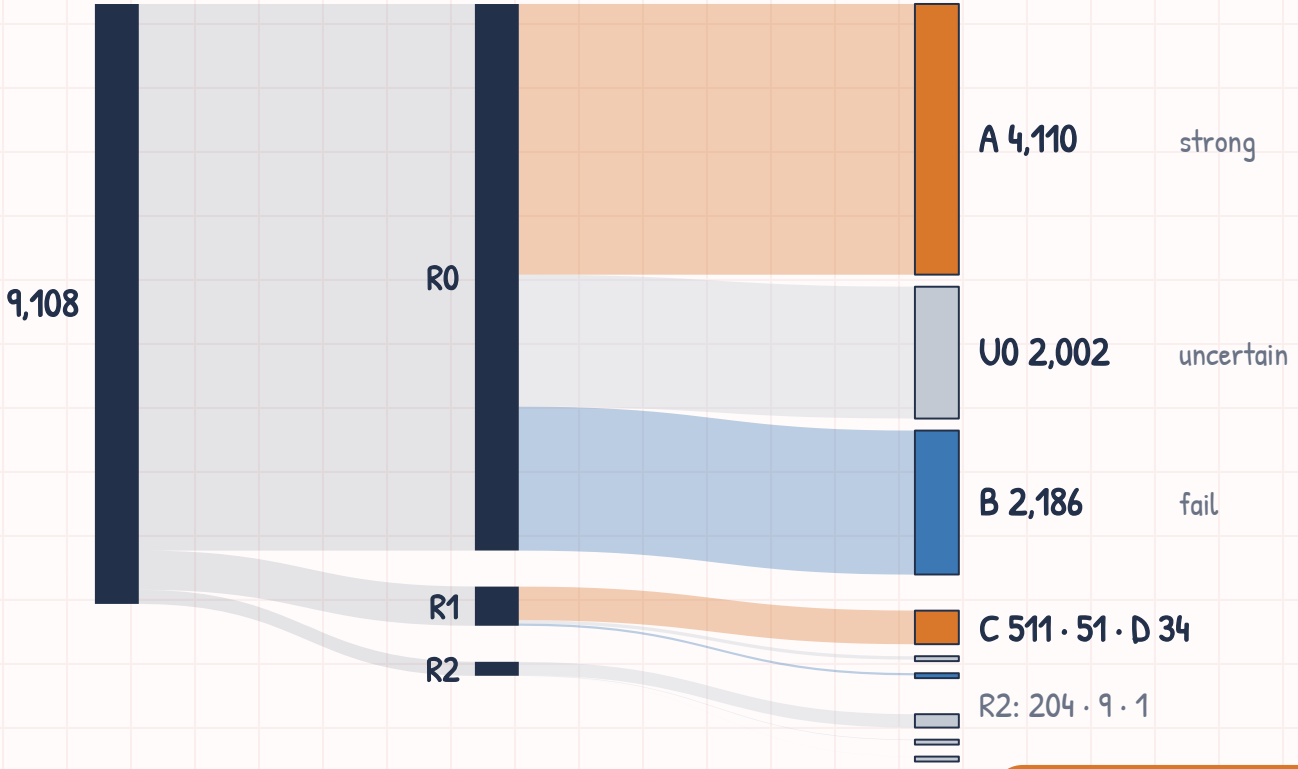
Probe



Groups and samples

example: cell C

1 papers → 2 open recall → 3 recognition → 4 cells → 5 samples



Stage 10 · M2 · M3

all papers

Stages 11–13 · X

593 target + 593 control

1,178 matched pairs

Stages 1–9

Pilot50 · Stage 8 (A, B)

target = A + C = 4,621 · control = B + D = 2,220

Cohort and behavioural labels

From the 15 Oct label package (9,108 papers) to a frozen 6,841-paper analysis cohort

Group	Recognition / open recall	Full	Pilot50
A	pass / R0 (no true author written)	4,110	20
B	fail / R0	2,186	20
C	pass / R1 (some authors matched)	511	7
D	fail / R1	34	3

- Pass = 4/4 option orders correct *and* mean true-option probability ≥ 0.5; fail = 0–1/4
- Pilot50: seed 35150, keeps the 20 Stage 4 papers + 30 new
- 20 A/B pairs matched on exact field and citation bin, then $|\Delta\text{year}| + 2 \cdot |\Delta\text{authors}|$

Why A vs B?

Both groups write **no** correct author. They differ only in whether the model recognises the true list. That isolates "knows but cannot say".

Caveat

In the 0–2-citation bin the strict pass rate is **25.5%**, above the planned ≤10% floor: "pass" is not proof of stored knowledge.

One fixed model, prompt and score

Everything later is measured under these settings

Model	Qwen/Qwen3-32B, BF16, revision 9216db57, 64 blocks, d = 5,120
Layers	L16, 24, 32, 40, 48, 56, 62 → target L63
Readouts	Public J-Lens · Vanilla (logit lens) · real final logits
Position	last input position; context limit 512
Names	encode(prompt) + encode(" " + author)
Hardware	29 blocks on GPU, 35 offloaded to CPU

> Who are the authors of the paper titled "{title}" ({year})?
> Authors:

Rank r = 1 + #tokens with a strictly larger logit

JMRR = mean over authors of mean over tokens of 1/r

Integrated = average of the 7 source layers (L63 excluded)

Guard: forced token IDs are checked, and a J-Lens/Vanilla final-logit gap > 0.1 rejects the run.

How groups and contrasts are built

labels: group lead's pipeline on the same Qwen3-32B · we use them unchanged

open answer -> R0 / R1 / R2

greedy, chat template

fuzzy name match: M, H, U

HR2 = (H+U) / (M+H+U)

R2: HR2 <= 0.5 · R0: M = 0 · R1: M > 0

a fixed rule, no human or model judge

multiple choice -> target / control

true list + 3 real teams of other papers

(same field, team size, citation bin, ~year)

4 orders · p(letter) from next token

target: 4/4 and mean p(true) >= 0.5

control: <= 1/4

5 false titles per paper

same field, own year kept

min-cost on |dyear|/5 + |dauthors|

+ |dtitle tokens| + 1.5 |dcitation bin|

no shared surname, Jaccard <= 0.6

79% strict tier

statistics

target vs control only within citation bin

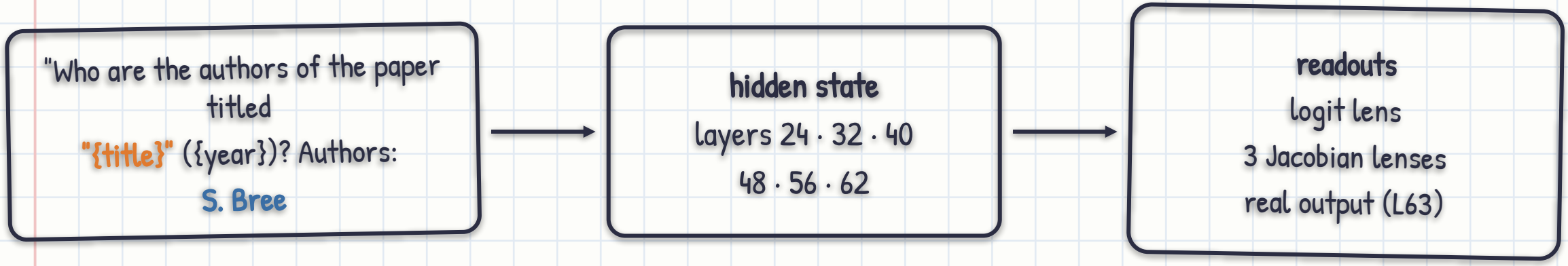
10 bins with >= 30 per group

weight n_t n_c / (n_t + n_c)

95% bootstrap over papers, 2,000

sensitivity: drop bins 0-4

How we read the model



Score = rank of the correct next name token,

as log10(1 / rank): 0 = rank 1, -1 = rank 10

7 title conditions: the true title, 5 matched false titles, no title

title-specific signal

= true title - mean(5 false titles)

Same name, same tokens: what is left is knowing

this paper

Lenses (model weights never change; Jacobians to L63 averaged over prompts): public = 557 WikiText-103 prompts (Neuronpedia) · self100 = 100 frozen WikiText-103 prompts · fitA = 100 other WikiText-103 prompts (Stage 9: one skipped prompt replaced -> 100 valid)

Can we trust the numbers?

fast = slow

One forward pass vs one per prefix: max gap 0.37 logit (bf16 rounding); ranks 1-41 identical

matches Stage 6

972 tokens: same token ids, median log-prob gap ~1e-5

negative control

Punctuation tokens show ~0 title-specific signal at every layer

Caveat: the parity tolerance was raised 0.25 -> 0.5 after seeing these bf16 gaps.

Line A

Is it about this paper?



small samples asked

Does the true title beat a crossed or shuffled title? Does A gain more than B?

→

the full cohort answered

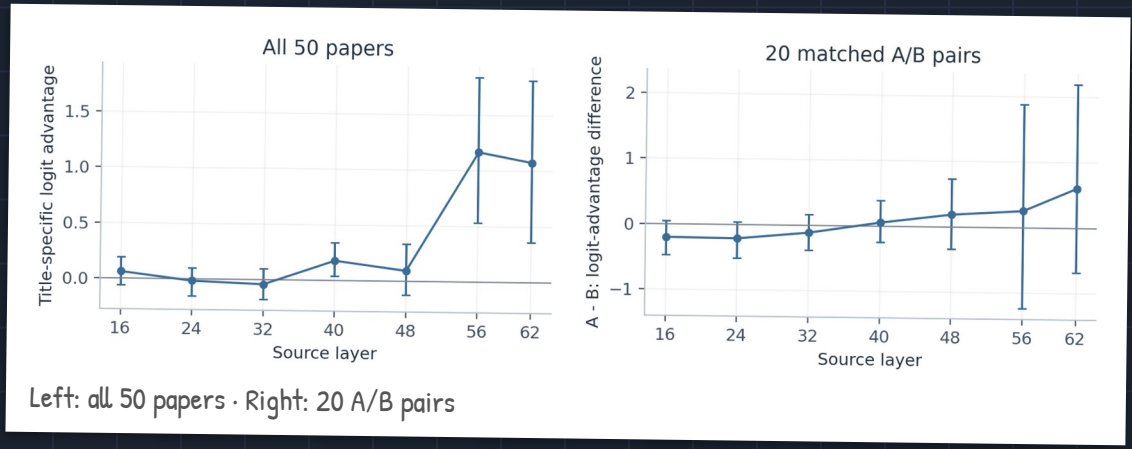
Title-specific signal for every token; target beats control only late; name prior split out; probe controls

Crossed title–author control

Is the gold name high because of the title, or because the name is easy?

Paper P (author G) and source paper Q (author F, same field & bin).
M = logit(G) − logit(F) · T = M(P title) − M(Q title)
A fixed preference for G cancels out.

- All 50 papers: L56 +1.17 [0.53, 1.84], L62 +1.08 [0.36, 1.82]
- Matched A–B, 7-layer integrated: +0.08 [−0.43, +0.65]

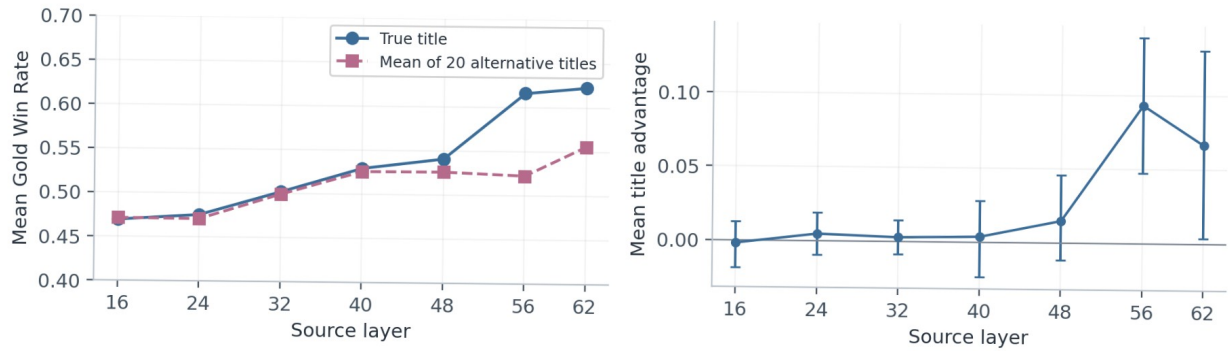


Left: all 50 papers · Right: 20 A/B pairs

The title changes which name wins in late layers. Whether A gains more than B: **not established**.

Shuffled titles: the true title helps

Same 100 candidates, 20 metadata-matched wrong titles per paper, 1,000 contexts; only the title changes



All 50 papers: true vs mean of 20 alternative titles

- Pooled L56: 0.616 vs 0.523 → +9.3 pp [4.7, 13.9], 35/50 papers positive
- Leave-one-out: +8.6 to +10.0 pp
- Vanilla L56 +10.6 pp, final output +7.2 pp
- Matched A–B integrated: +0.018 [−0.021, +0.063]

Two different questions: "does the title help?" **yes**; "does it help A more than B?" **not shown**.

Does the title effect survive the obvious objections?

Source-author overlap

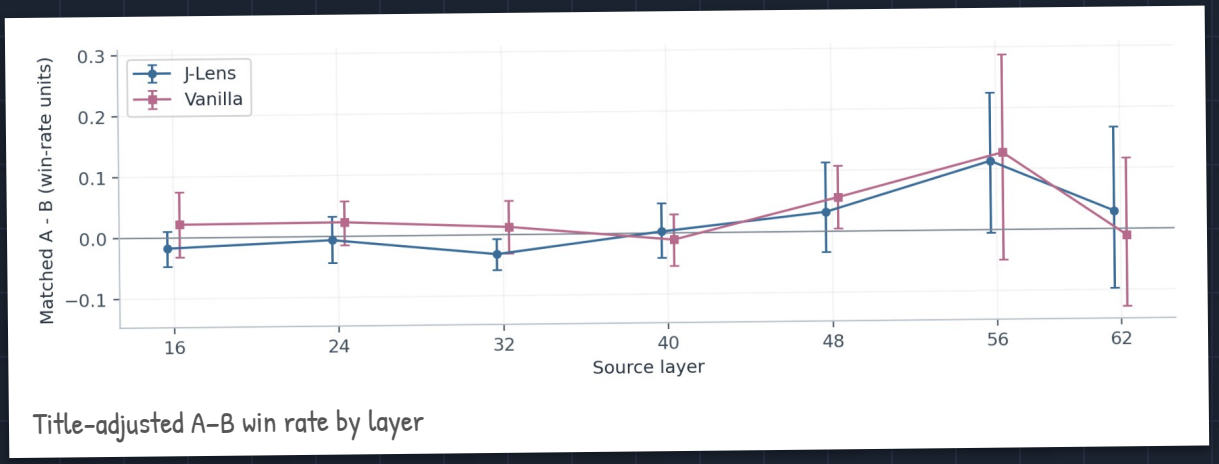
199/1,000 wrong titles come from a paper whose author is in the false pool. Excluding them: +0.026 → +0.028.

Candidate weighting

Author-, unique-token- and non-collision weights: pooled +0.026 to +0.032; all six A–B CIs include 0.

Low citations

0–2 citations (n = 6): L56 +0.22, 5/6 positive, but 0/6 beyond its own 20-title tail (≤ 0.05).



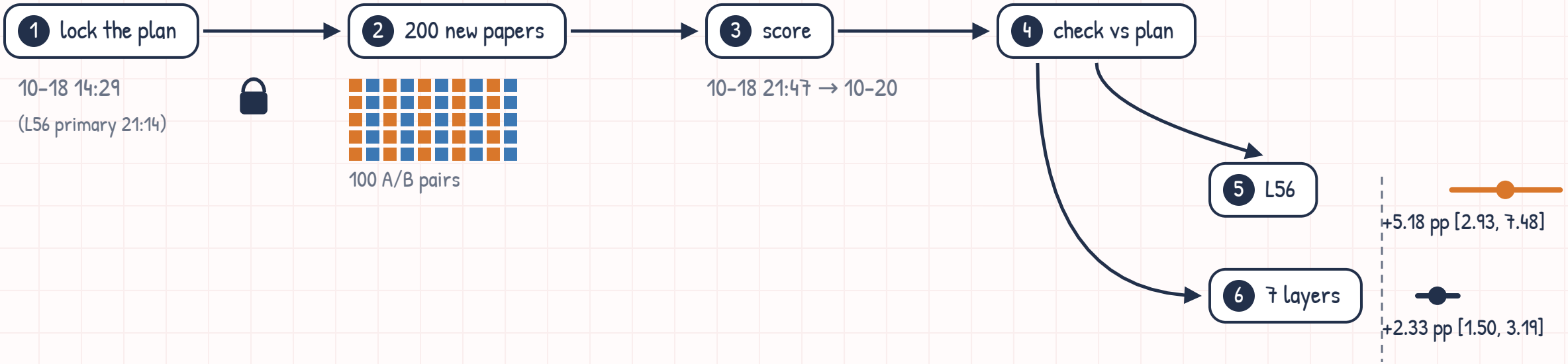
Title-adjusted A–B win rate by layer

Layer heterogeneity

L56 +0.11 but L32 **−0.03** (CI below 0). Reporting only the positive layers would mislead.

Independent replication (Stage 8)

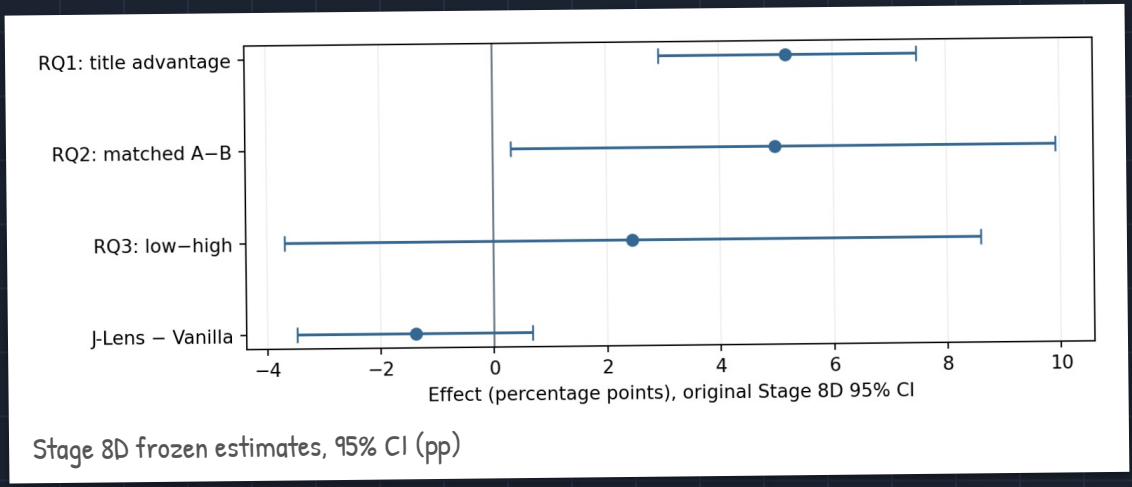
plan first, data second



RQ1 = win rate(true title) − mean win rate(20 false titles)

first author vs 100 false names · J-Lens · paper bootstrap

On 200 new papers the title effect replicates



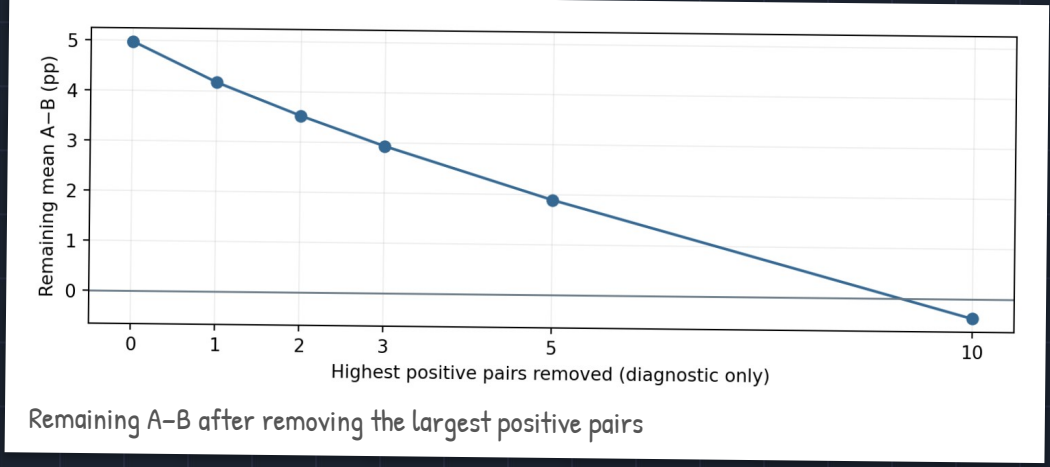
Stage 8D frozen estimates, 95% CI (pp)

	Estimate	95% CI
RQ1 title	+5.18 pp	[+2.93, +7.48]
RQ2 A–B	+4.98 pp	[+0.32, +9.91]
RQ3 low–high	+2.45 pp	[−3.69, +8.60]
J-Lens–Vanilla	−1.38 pp	[−3.48, +0.68]

Citation bands: low +6.8 [2.6, 11.6], mid +4.8, high +4.4 [−1.1, 10.0]

RQ1 is the first **independent** confirmation: correct titles help the gold author on new papers. J-Lens still adds nothing over Vanilla.

But the A–B gap rests on a few pairs



Remaining A–B after removing the largest positive pairs

- 100 pairs: 55 positive, 45 negative; median +4.4 pp
- One pair (S8P094) ≈ +84 pp
- Remove top 5 → +1.9 pp; top 10 → −0.4 pp

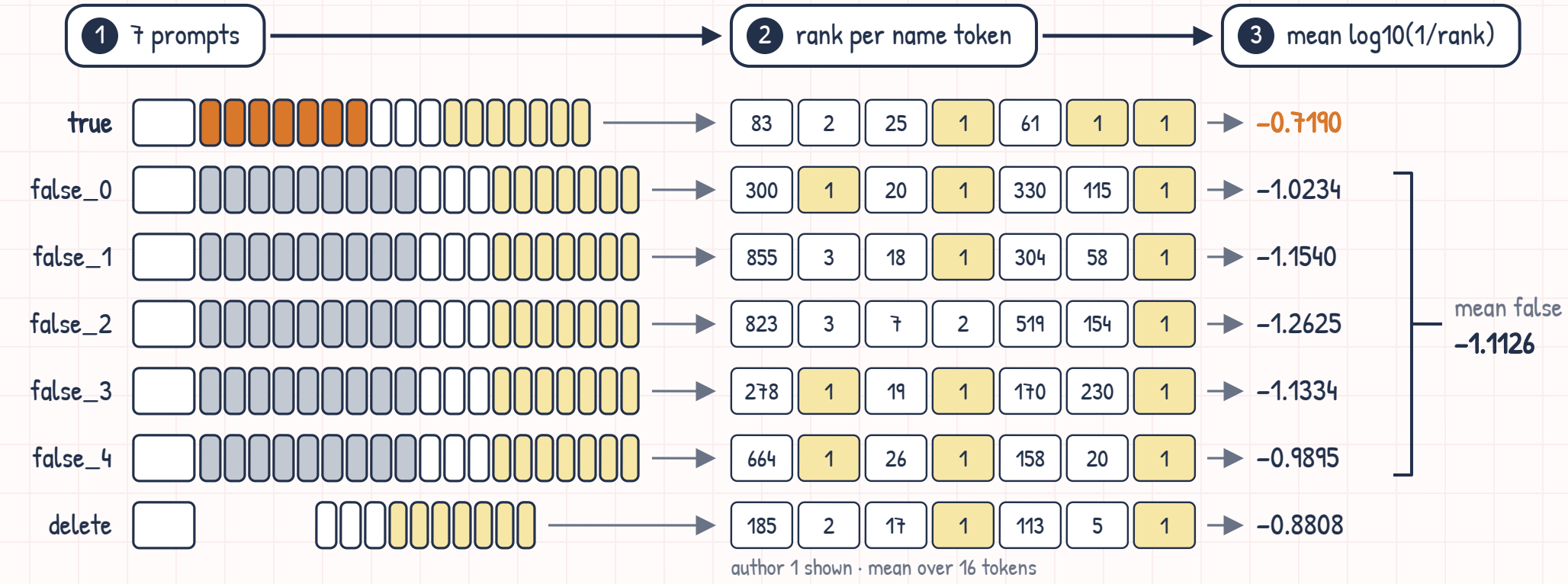
8E external audit (post-hoc)

- Unique-token weights: RQ1 +4.6, RQ2 +5.4 pp
- Wrong-title length gap: +0.12 pp per token (association only)
- Scripts not yet archived: not a completed experiment

Small samples can show a positive A–B mean that a handful of papers carry. This is the main reason Stage 10 moved to the full cohort.

Scoring: rank → log → spec

Coifman, Bonanno, Rafaeli (2007) · output layer



4 contrast

spec = mean nlr(true) − mean over 5 false titles

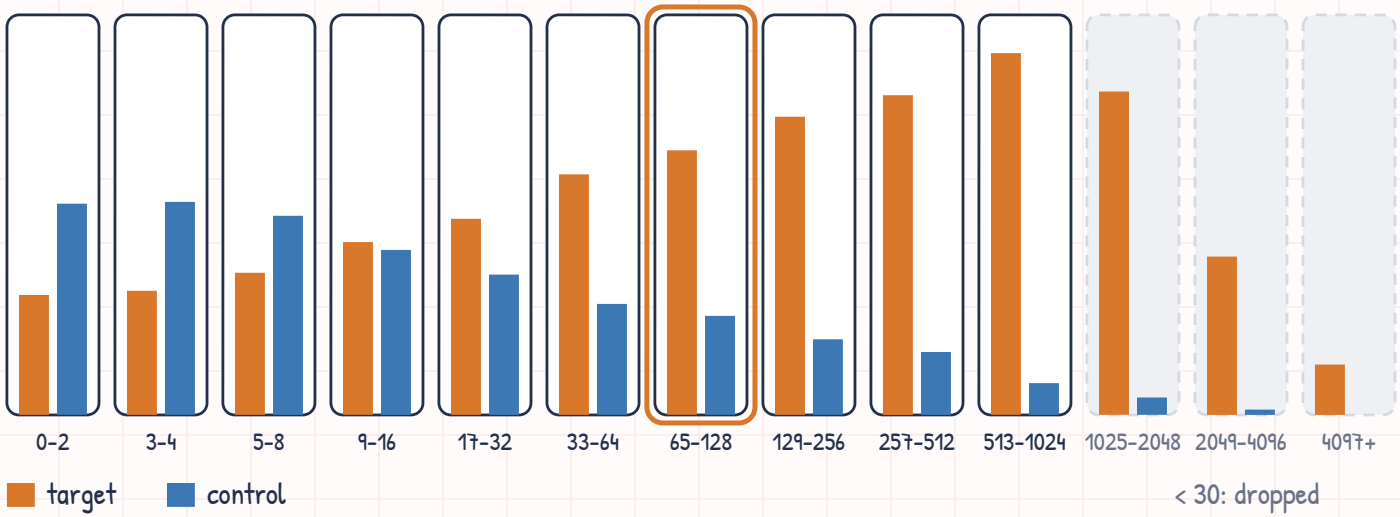
= −0.7190 − (−1.1126) = +0.3935

total = true − delete = +0.1618 · topic = mean false − delete = −0.2318

Statistical conventions

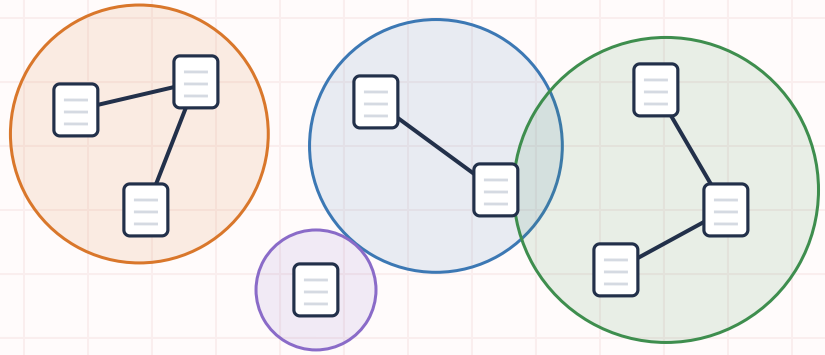
example paper: bin 65–128

1 citation bins → 2 target − control → 3 weighted mean



$$\Delta = \Sigma\, w_b \cdot (\text{mean}_t - \text{mean}_c), \quad w_b \propto n_t \cdot n_c / (n_t + n_c)$$

4 teams → 5 resample teams



shared author name = one team

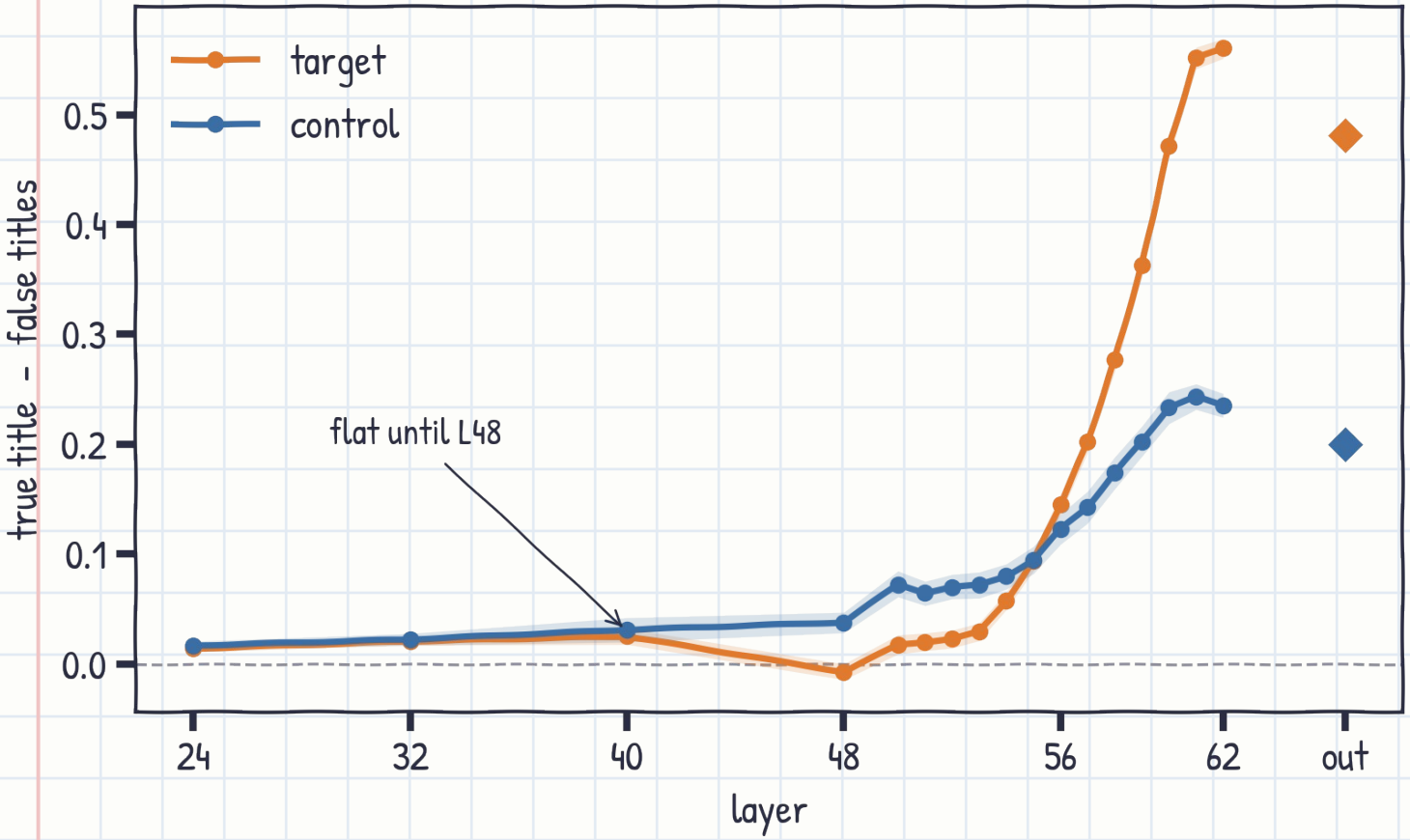
7,247 teams



one team, one fold · 2,000 draws

Knowing the paper shows up only at the end

title-specific signal · public Jacobian lens (lines) and model output (diamonds)



L24–L48: ~0 for both groups

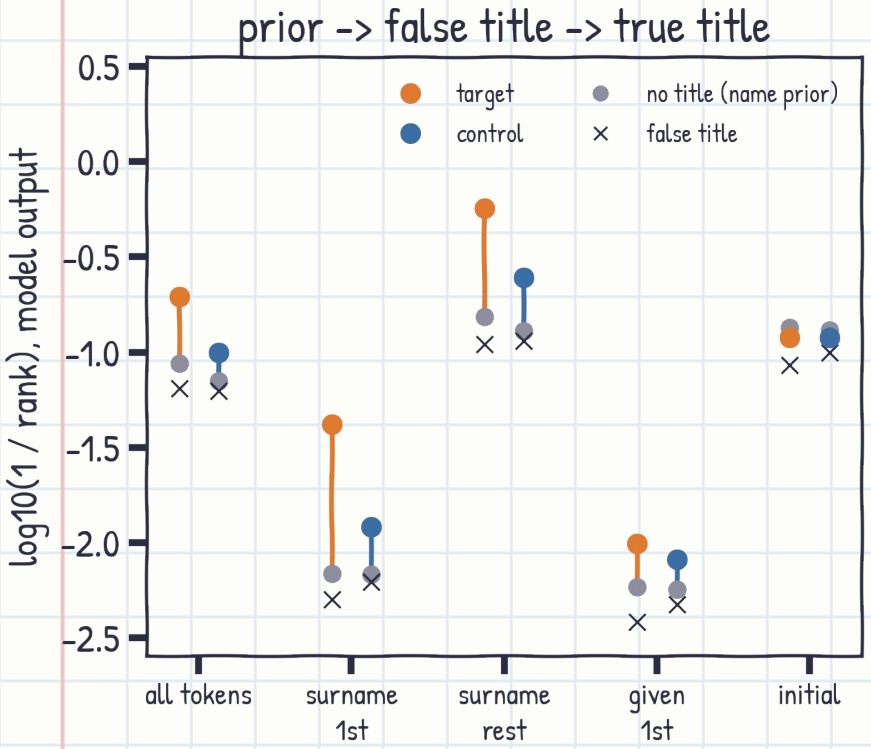
rises at **L56**

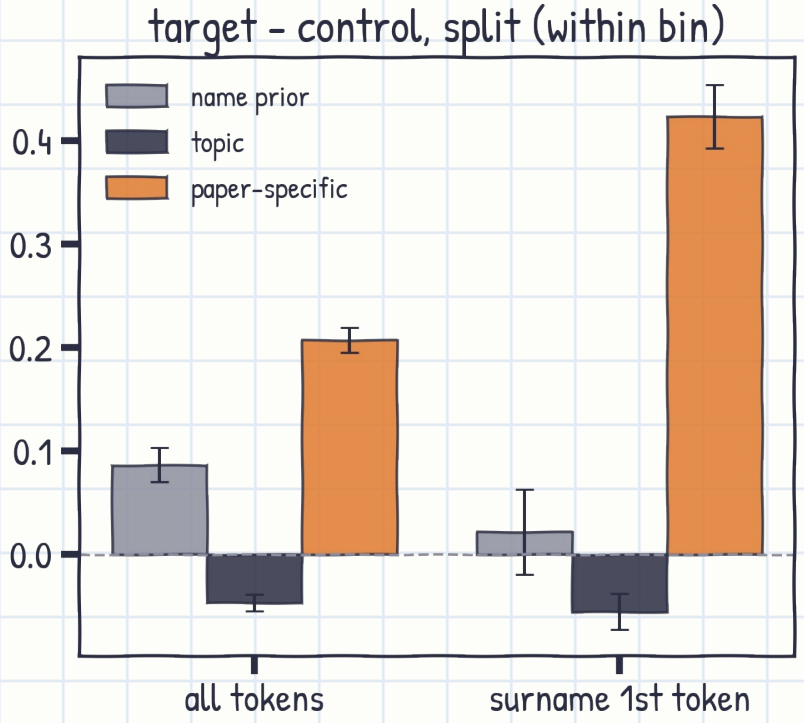
L62: target +0.56 vs control +0.24

Stage 11

How much is just the name prior?

model output · no title vs false title vs true title · all 9,108 papers





no title: rank ~12

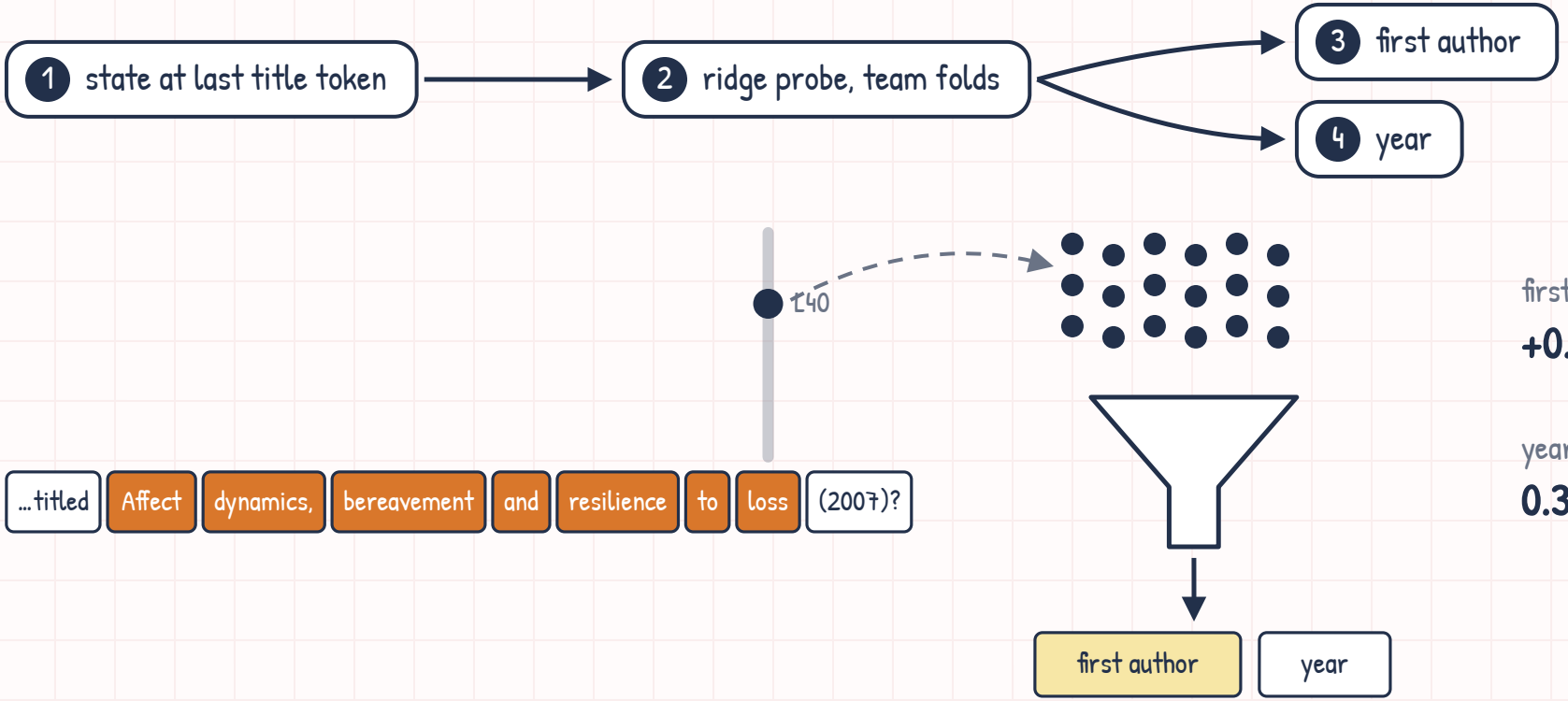
target - control: +0.09 prior, +0.21 paper

surname 1st: prior ~0, paper +0.42

prior vs recognition rho **0.09**

Probe

Coifman, Bonanno, Rafaeli (2007) · state at "loss"



first author, L40

+0.061

year R^2, L16 → L52

0.38 → 0.56



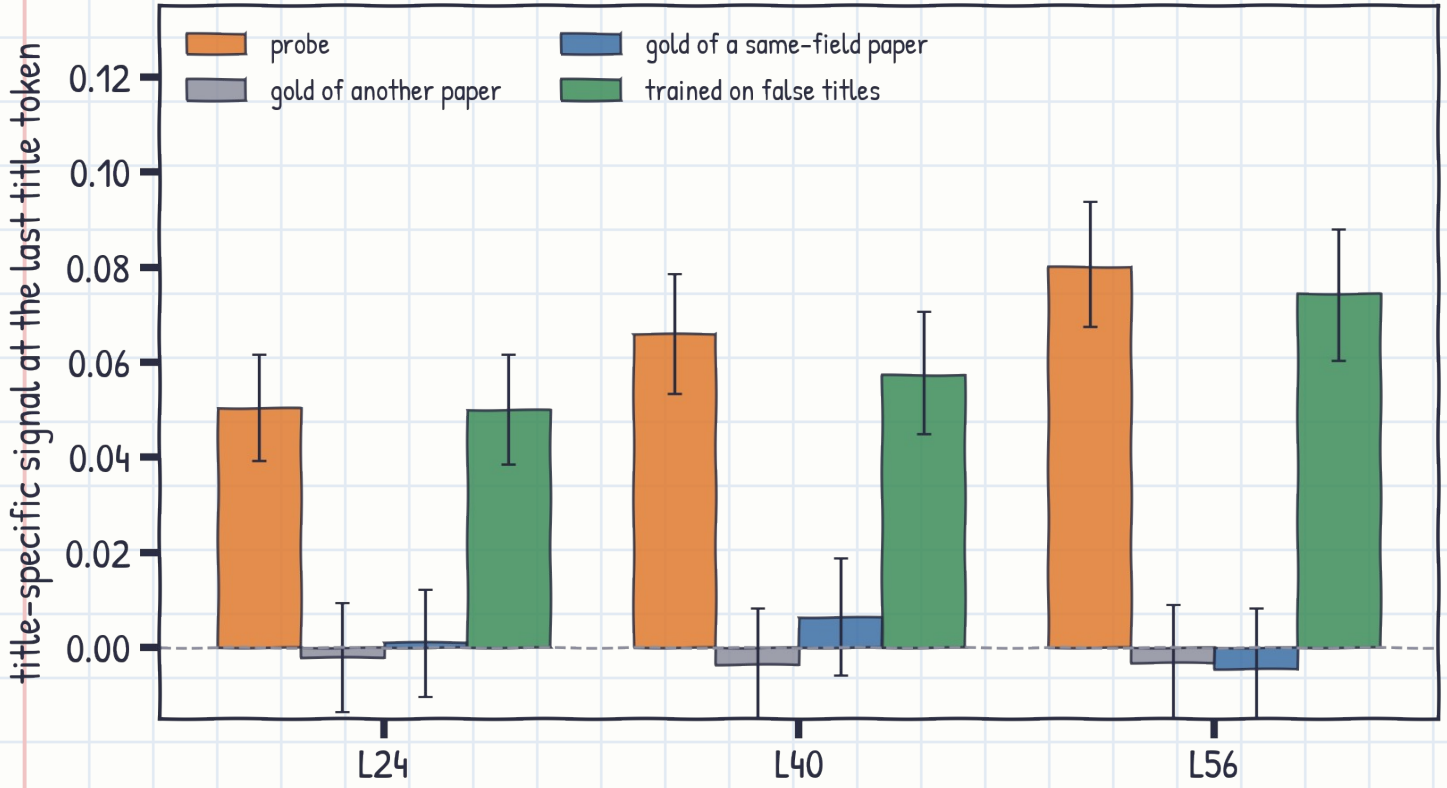
colour = author team; one team, one fold

spec = probe(true title) − probe(false title)

probe score = log10(1/rank) of the true first-author token

The probe signal is paper-specific

last title token · 5-fold ridge probe · re-checked: author-team folds, title reuse removed



another paper's authors: 0

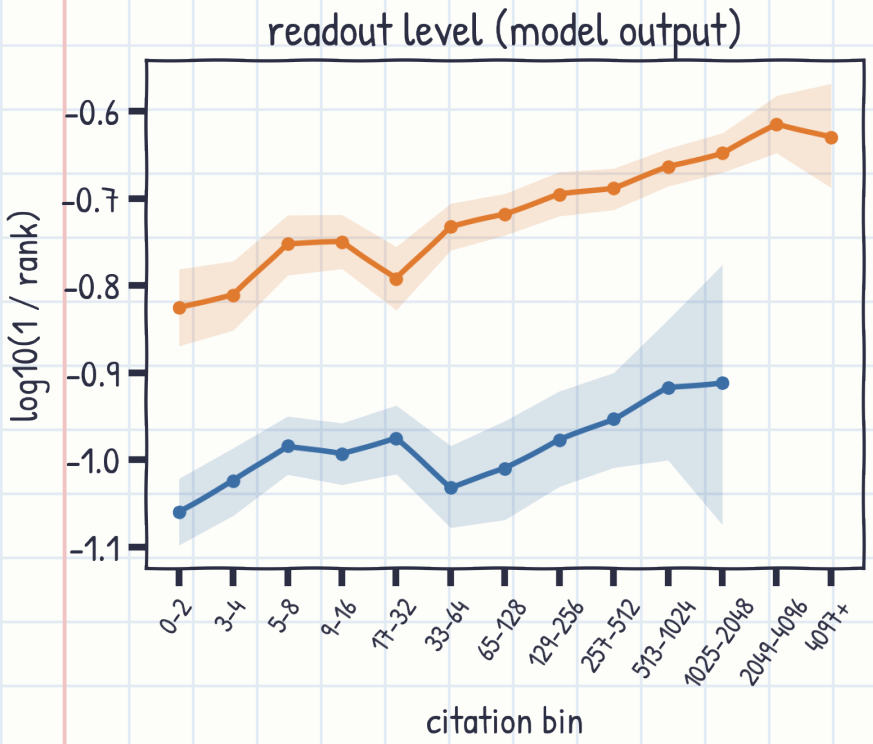
same-field paper: 0

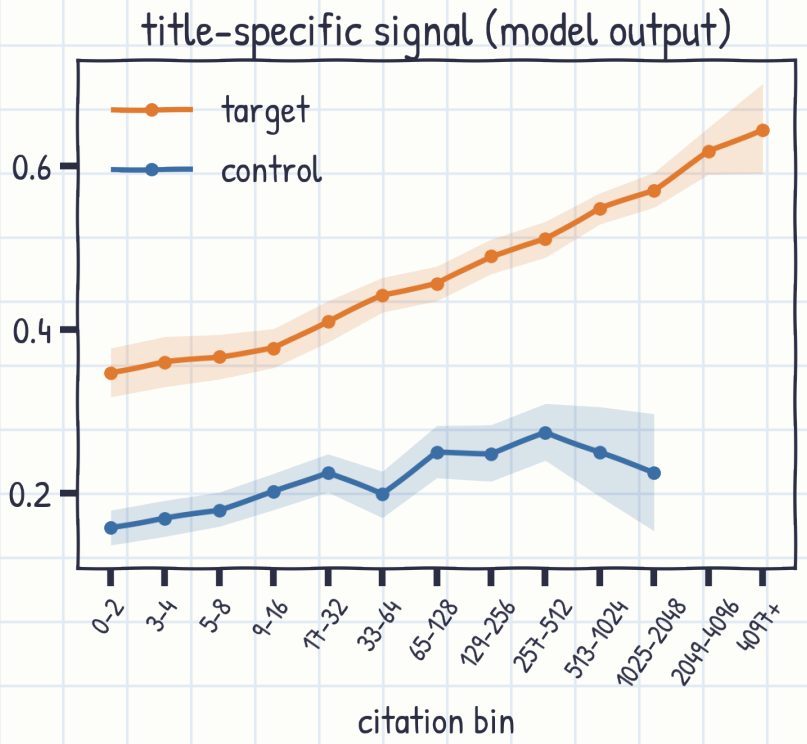
so it is this paper, already at L24

lenses cannot read it

More citations, more paper knowledge

model output · bins with < 10 papers omitted





Target: +0.35 -> +0.64

Spearman (title-specific)

0.45 all · 0.33 target · 0.19 control

Target > control in every bin

Group comparisons

example paper: cell C

1. 9 cells
2. P1: A – B
3. P2: C – A

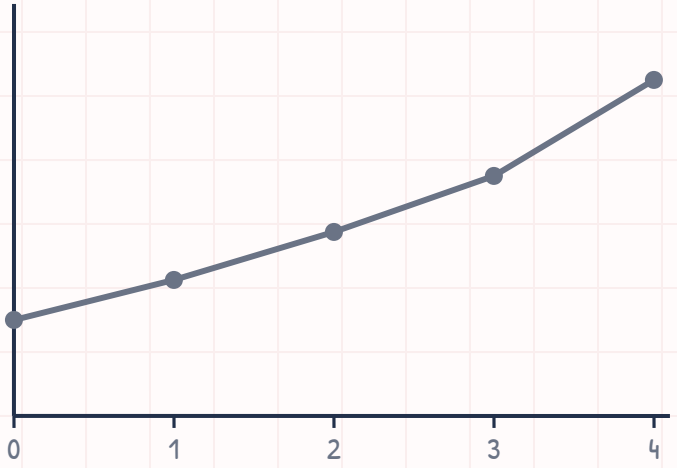
4. dose: orders right

P1

P2	strong	uncertain	fail
R0	A 4,110	U0 2,002	B 2,186
R1	C 511	51	D 34
R2	204	9	1

P1 Δ_1

P2 Δ_2



schematic, not results · 示意，不是结果

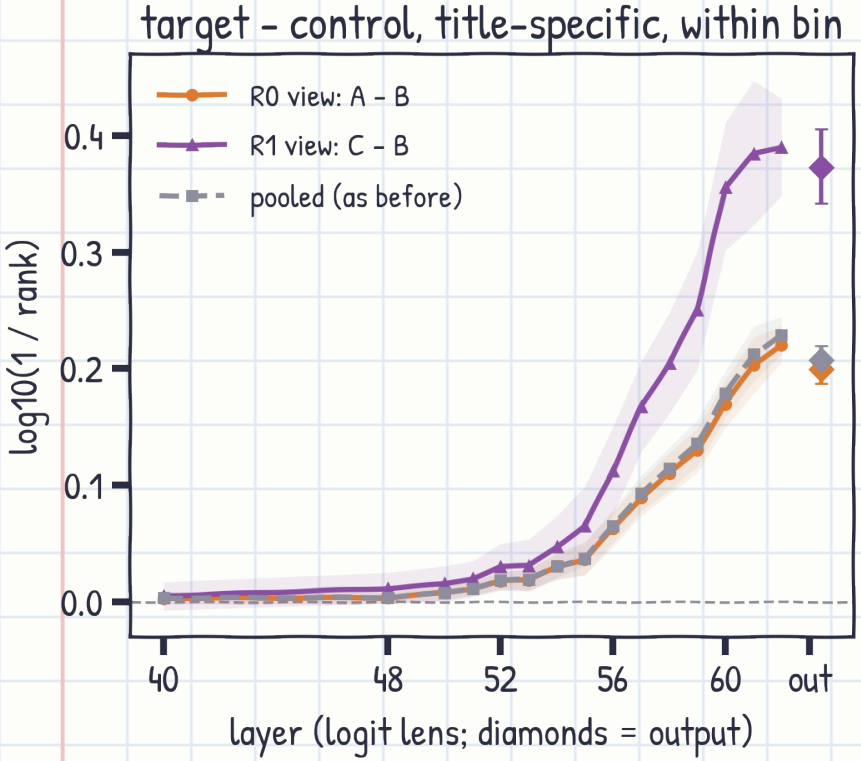
contrast = Σ w_b (mean_cell1 – mean_cell2)

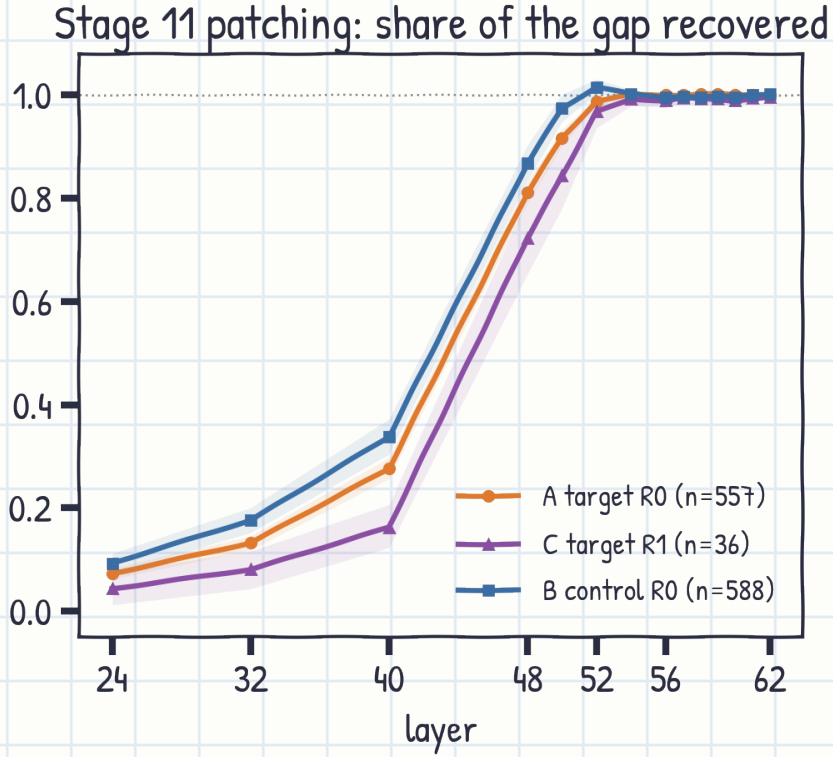
title-specific signal, unwritten authors, output layer; team bootstrap

Stage 10–11

Is it just the R1 papers? No.

R0 = no true author written · R1 = a few written · A/B/C/D = target/control x R0/R1





R0 only (A - B):

output +0.199 vs +0.207, same shape

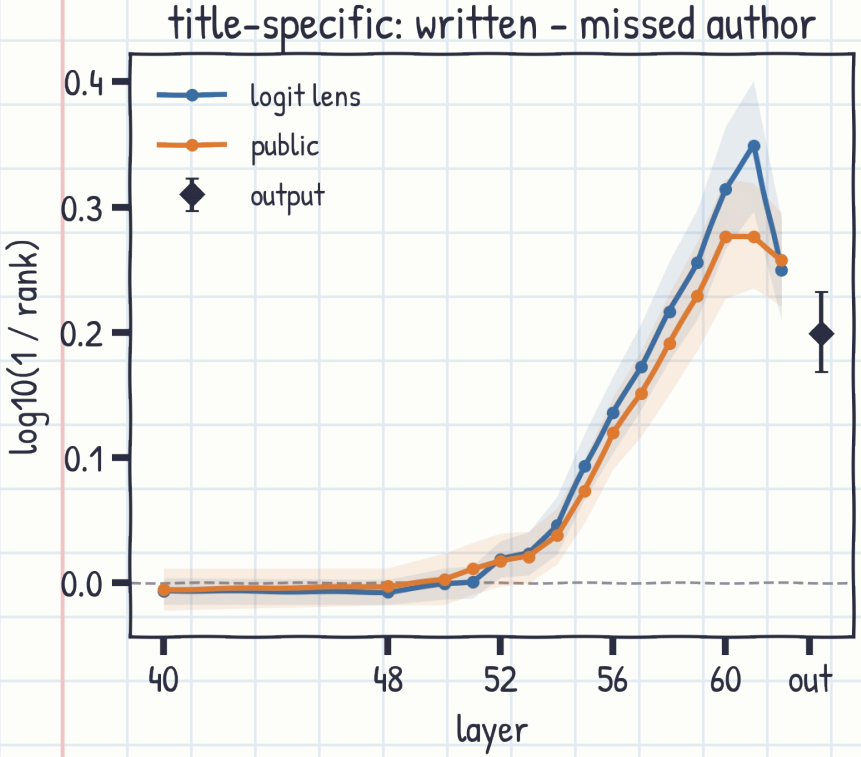
R1 (C - B): +0.37, also late

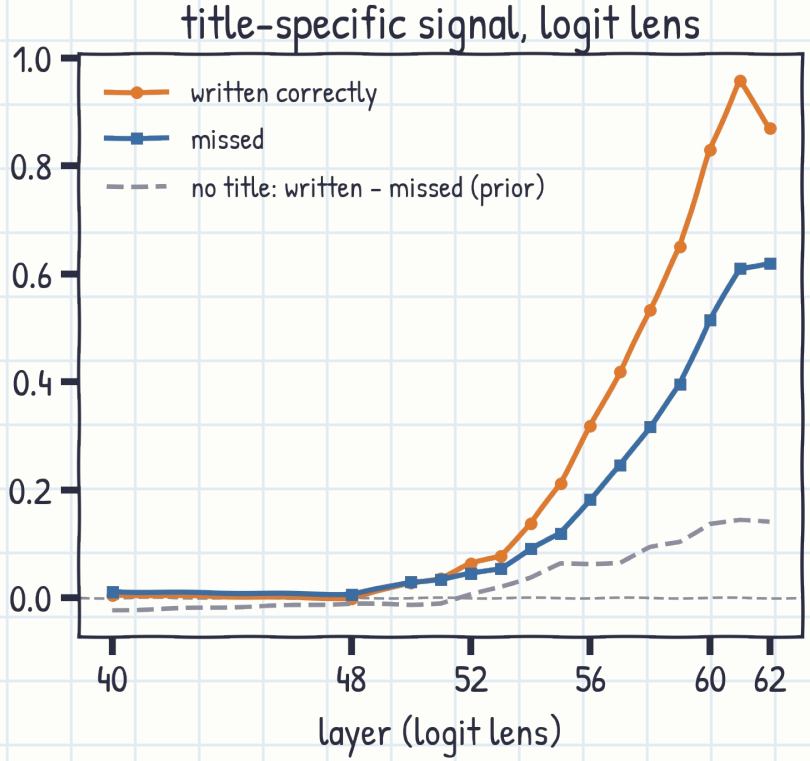
C - A within target: +0.11, from L56

patching: R1 a bit later, done by L52

Authors it wrote vs authors it missed

inside 495 R1 papers · paired within paper · 95% CI





written − missed, output: +0.20

~0 up to L52, grows from **L54**

no title (prior): +0.10

same without first authors

Stages 1–9 · 20–200 papers

What we saw

- True title helps: pilot L56 +9.3 pp; 200 new papers +5.2 pp
- A–B uncertain in the pilot; +5.0 pp on 200 papers but gone after dropping 10 pairs
- Low – high citation: inconclusive

Stages 10–11 · 9,108 papers

What scale showed

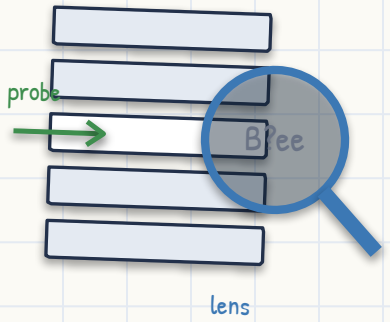
- Title-specific signal clear in every group, ≈0.44 at L62
- target – control ≈0 in middle layers, +0.21 [0.20, 0.22] at the output
- Name prior is only ~1/3 of the gap; ~0 on surnames
- Signal rises steadily with citations ($\rho = 0.45$)
- Same groups as the pilot (R0 only, A – B): +0.20 [0.19, 0.21], same late shape
- R1 papers (C) know more (+0.11 over A) but are just as late

Line A: from "the title helps" to "this paper, late, graded by fame"

The pilot was right that the title matters. It could not see that the target advantage is real but confined to the last layers, so the 7-layer average and the L56 endpoint both diluted it. Restricting the full cohort to the pilot's A/B groups (R0 only) changes nothing: the R1 papers do not drive the result.

Line B

What can each readout see?



small samples asked

Is J-Lens better than Vanilla? Is the fitted lens stable?

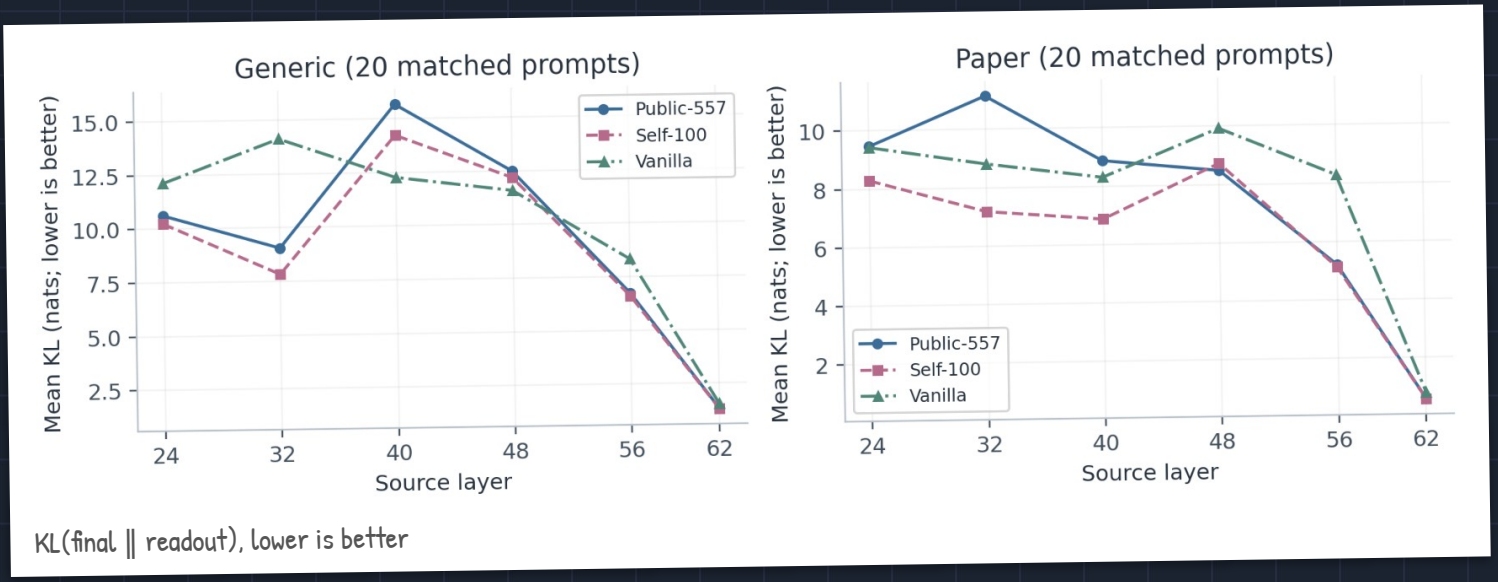
→

the full cohort answered

Lenses read name shape earlier, not paper identity; a probe at the title token reads paper-specific information

Two lenses: Public-557 vs Self-100

Can a middle layer approximate the final next-token distribution? (20 generic + 20 paper prompts)



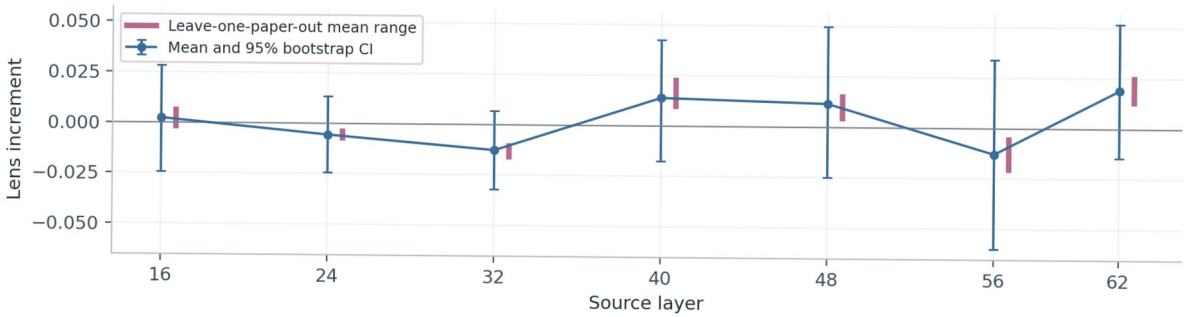
KL(final ‖ readout), lower is better

- Both checkpoints numerically valid (63/63 and 6/6 finite)
- Self-100 has lower KL on 6/6 generic and 5/6 paper layers (paper L32: 11.1 → 7.2)
- But at paper L62, Self-100 KL beats Vanilla (0.58 vs 0.81) while Top-10 overlap is worse (0.36 vs 0.72)
- Author pilot: both lenses 14/20 full-name wins at L48

Lesson: a better distribution match does not mean better author ranking. Fidelity cannot stand in for the task.

Cross-stage audit: does J-Lens add value?

Read-only joins of Stages 5–7.3; 20,000 bootstrap draws



J-Lens minus Vanilla title advantage, with leave-one-out ranges

Stage 5 vs 7.3 at L56 r = 0.45, ρ = 0.47; signs agree 32/50

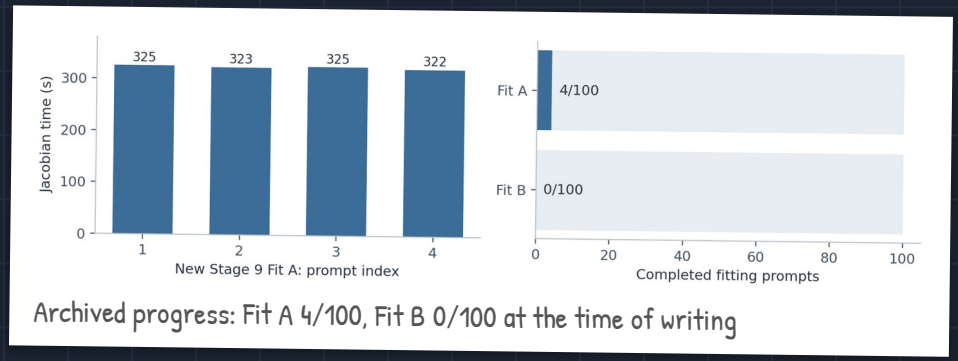
J-Lens − Vanilla	+0.002 [−0.008, +0.012]; median −0.002; 24/50 positive
A−B of that increment	−0.014 [−0.033, +0.004]

Dropping one paper can flip the sign of the J-Lens increment. The pilot shows **no added title-specific value** from J-Lens over Vanilla. Everything so far is exploratory: L56 was inspected repeatedly.

Stage 9: is the lens stable across fitting data?

New FullFit protocol, different from the historical 50+50 split

	Historical split-half	Stage 9 FullFit
Fit data	Self-100 split 50 + 50	two disjoint 100-prompt sets
Holdout	20 generic + 20 paper	independent 100 WikiText
Layers	6	7 (adds L16)
Status	complete	only Fit A completed; Fit B and holdout not run



Archived progress: Fit A 4/100, Fit B 0/100 at the time of writing

- WikiText-103 train, 100–1,200 chars, deduplicated by hash, seed 3519001
- ~324 s Jacobian time per prompt, ~62 GiB CPU RAM; per-prompt checkpoints

Only Fit A was finished, and Stage 10 used it as "fitA". Without Fit B and the holdout, fitting stability is **untested**.

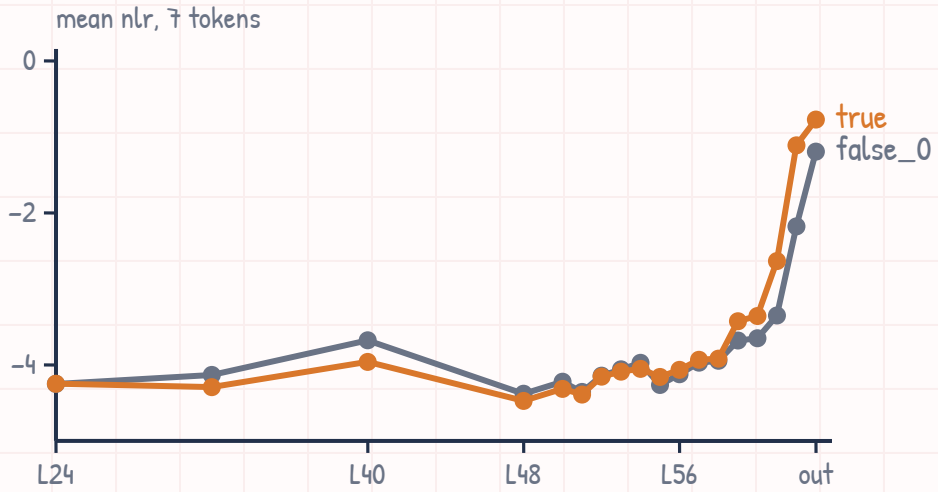
Layer-by-layer readout

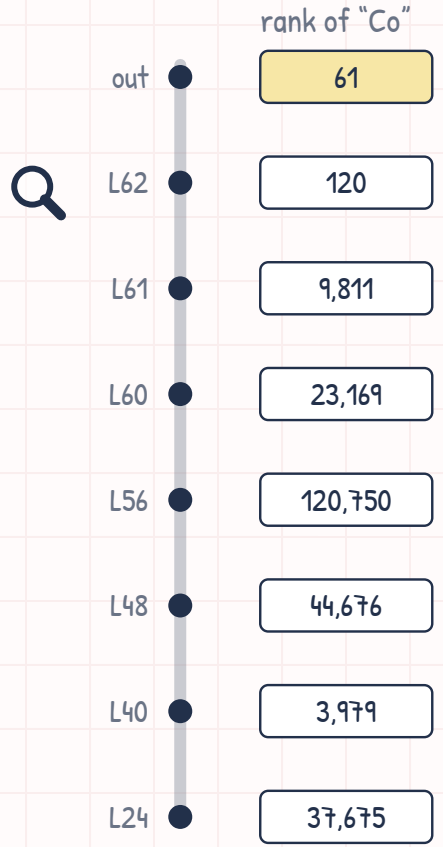
Coifman, Bonanno, Rafaeli (2007) · logit lens

1 prompt → 2 state at name → 3 lens → vocabulary → 4 rank of true token → 5 mean

nlr = log10(1 / rank)

rank = 1 + #{tokens scored above the true one}

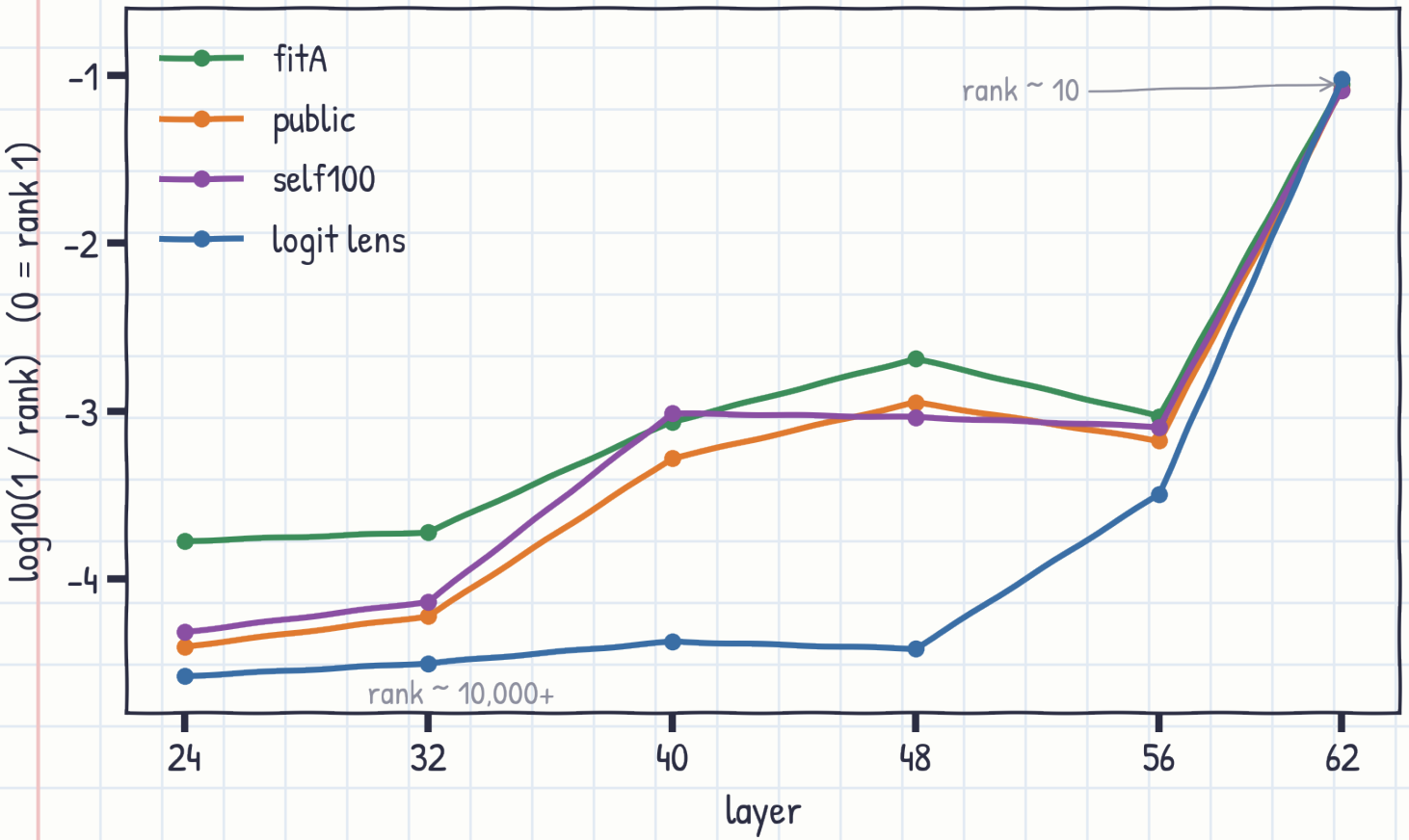




…titled	Affect	dynamics,	…	(2007)?	Authors	:	Kar	in	G	.	Co	if	man

Names become readable late

all 9,108 papers · true title



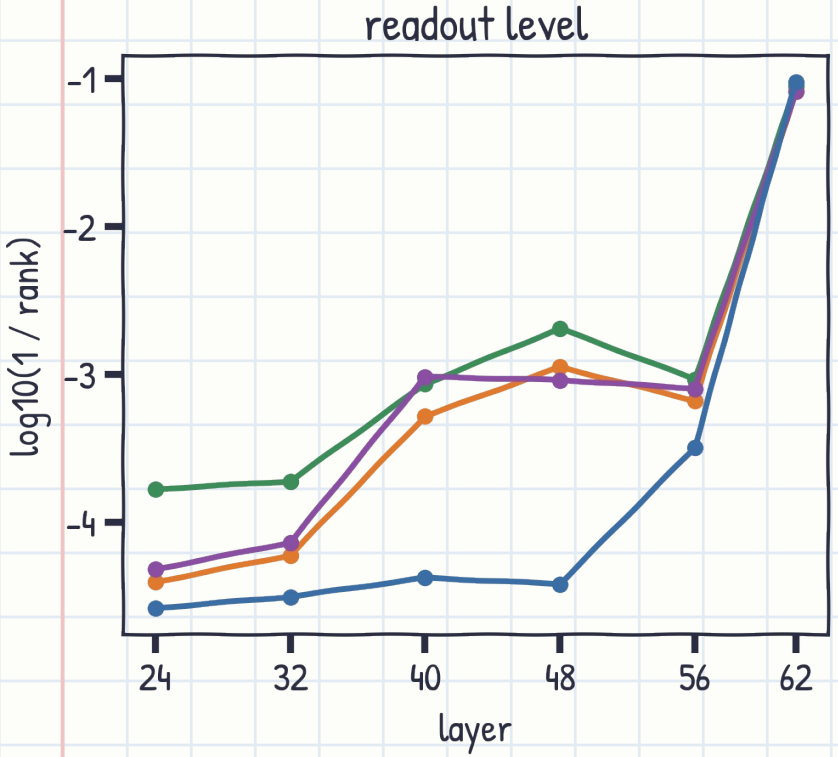
Logit lens: almost nothing until L48

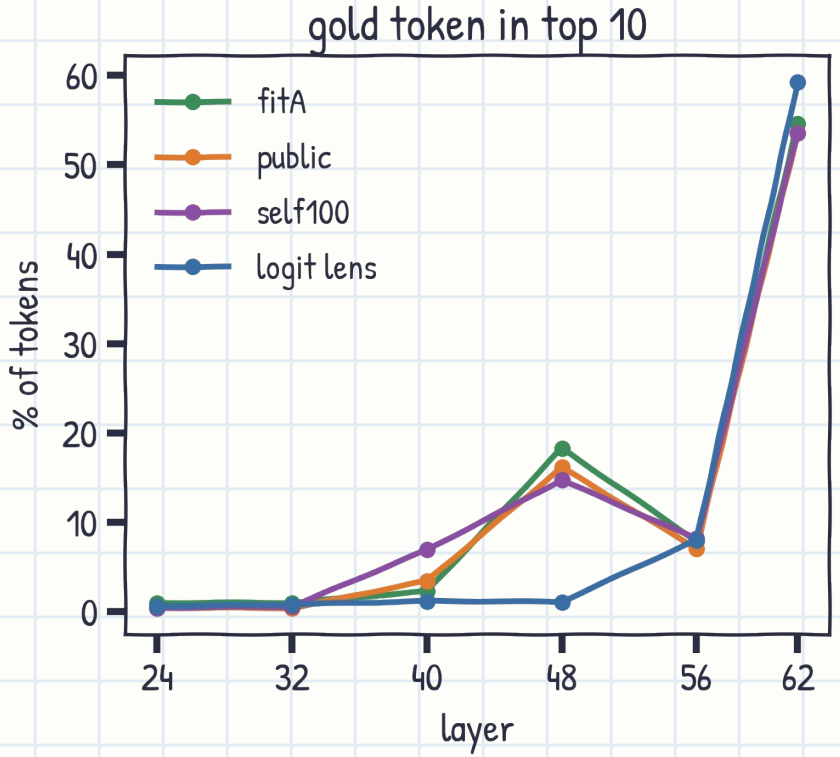
Jacobian lenses read more from L40

Everything jumps at L62 (rank ~10)

Which lens reads best?

all papers · true title





Middle layers:

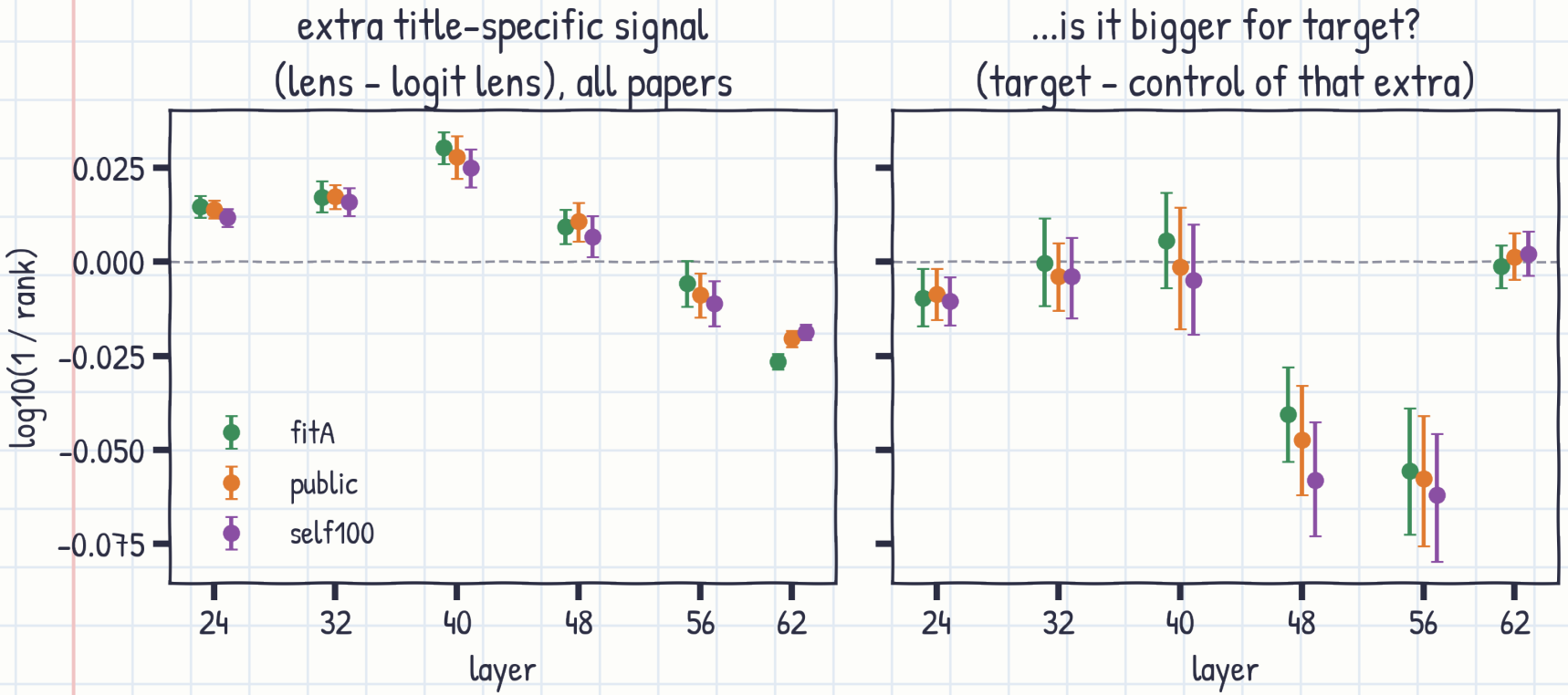
fitA best; at L48 15–18% top-10 vs 1% for logit

L62: logit lens best (59%)

Stage 10

Do lenses reveal hidden paper knowledge?

paired differences, 95% CI



Extra signal: **+0.01 to +0.03**

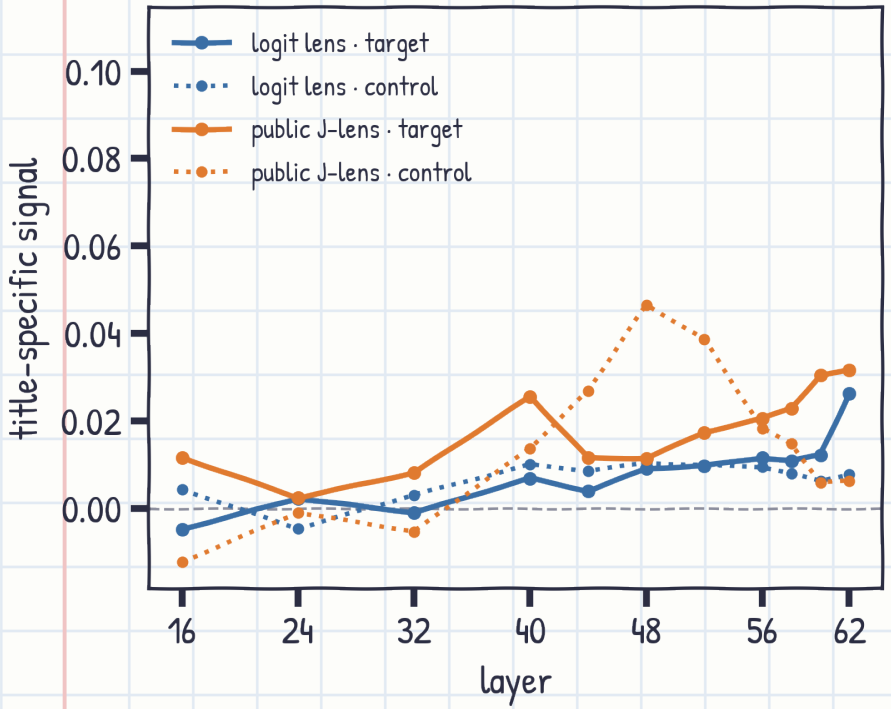
vs ~0.44 at L62

Not bigger for target
(even negative at L48–56)

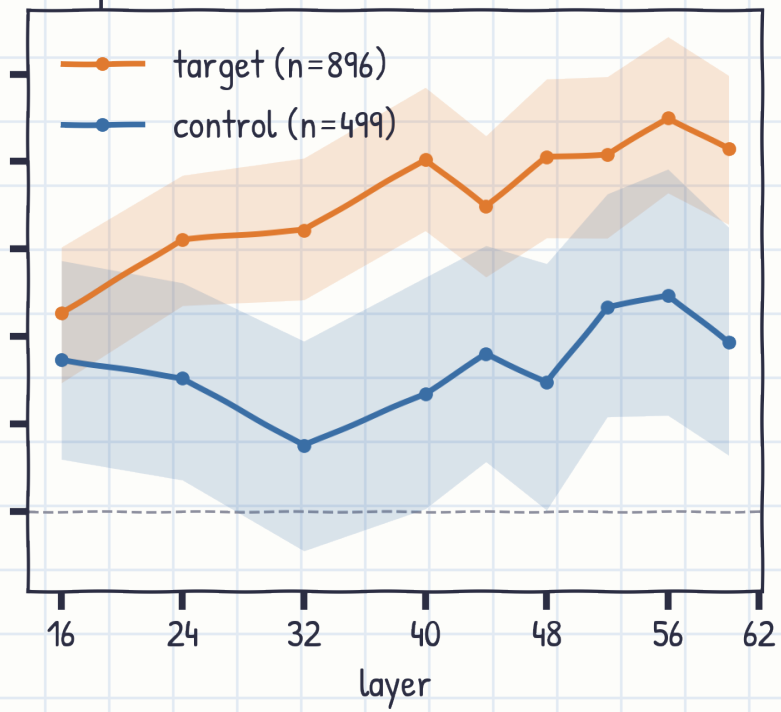
Read early: a probe finds it at the title

title-specific signal at the last title token · lenses: all 9,108 papers · probe: re-checked, title reuse removed: 1,863 papers (896 target, 499 control)

lens readout at the last title token



linear probe at the last title token (re-checked)



lenses: ~0.01–0.03

probe: 0.05 -> 0.08

target > control by +0.04–0.08

re-checked: team folds, title reuse removed

scores the first name token only (mostly an initial); subtracts false_0 only

read early (title, from L24), spoken late (names, L57–61)

Stages 1–9 · 20–200 papers

What we saw

- Self-100 fits the final distribution better; author ranking not better
- J-Lens − Vanilla ≈0 (7.4: +0.002; Stage 8: −1.4 pp)
- Stage 9: only Fit A fitted, stability not tested

Stages 10–11 · 9,108 papers

What scale showed

- Jacobian lenses: top-10 rate 15–18% at L48 vs 1% for logit lens
- Extra title-specific signal only +0.01–0.03, not target-specific
- At L62 the plain logit lens is best
- Linear probe at the title token: +0.05 → +0.08 (re-checked), paper-specific

Line B: the information is there, but not in a form the lenses read

Both phases agree: no lens reveals hidden paper knowledge in middle layers. Scale adds the positive side: a trained probe can decode it at the title, so the limit is the readout, not the absence of information.

Line C

When does it appear, and when is it used?



small samples asked

Which layer peaks? (L56 kept standing out)

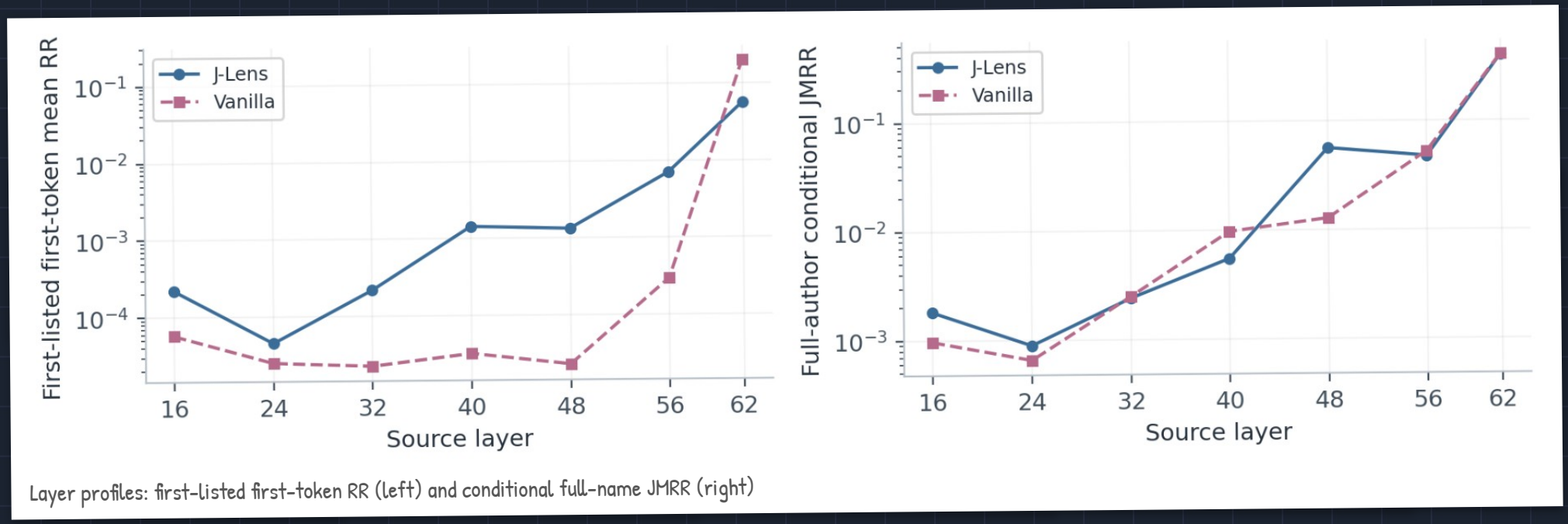
→

the full cohort answered

Readable late (L57–61) but read from the title early (L24–50)

First readout: is any author signal there?

20 papers, 93 authors (A8 / B8 / C3 / D1); ~91 s per paper, ~29.5 GiB GPU



Layer profiles: first-listed first-token RR (left) and conditional full-name JMRR (right)

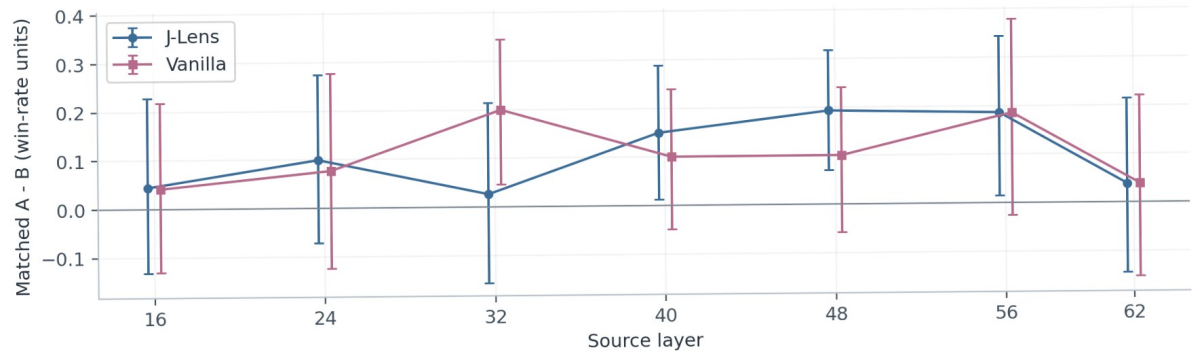
- The pipeline works and J-Lens rises above Vanilla in middle layers
- L56 first-author mean RR 0.0070, median only 0.0009

One paper = 71% of the L56 mean

Dropping it: 0.0070 → 0.0021. A layer peak can be a single easy paper, so every later stage adds controls.

Gold vs 100 real false authors

50 matched (same field, token count, year) + 50 random names per paper; 75,000 scored pairs



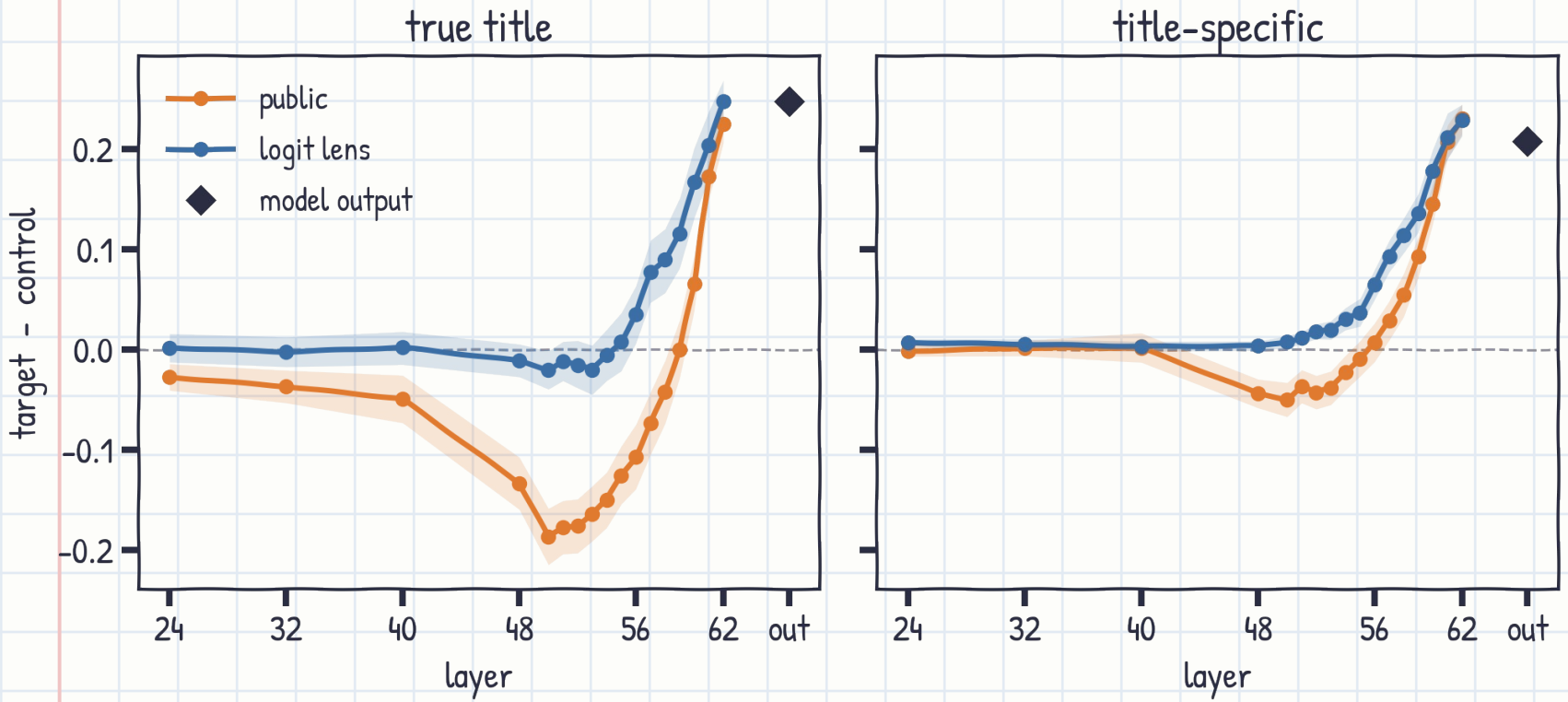
Matched A–B gold win rate, 100-candidate pool

- J-Lens A–B: L40 +0.15, L48 +0.19, L56 +0.19 (CIs above 0); L62 +0.04 (spans 0)
- 7-layer integrated: +0.105 [−0.012, +0.228]
- 156/5,000 false names share the gold first token (ties)
- 100 names → only ~46 distinct first tokens

Gold beats real names in middle layers for A more than B. But the same names might win under any title: 7.3 tests that.

Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI



Middle layers: **no head start**

L62 / output:

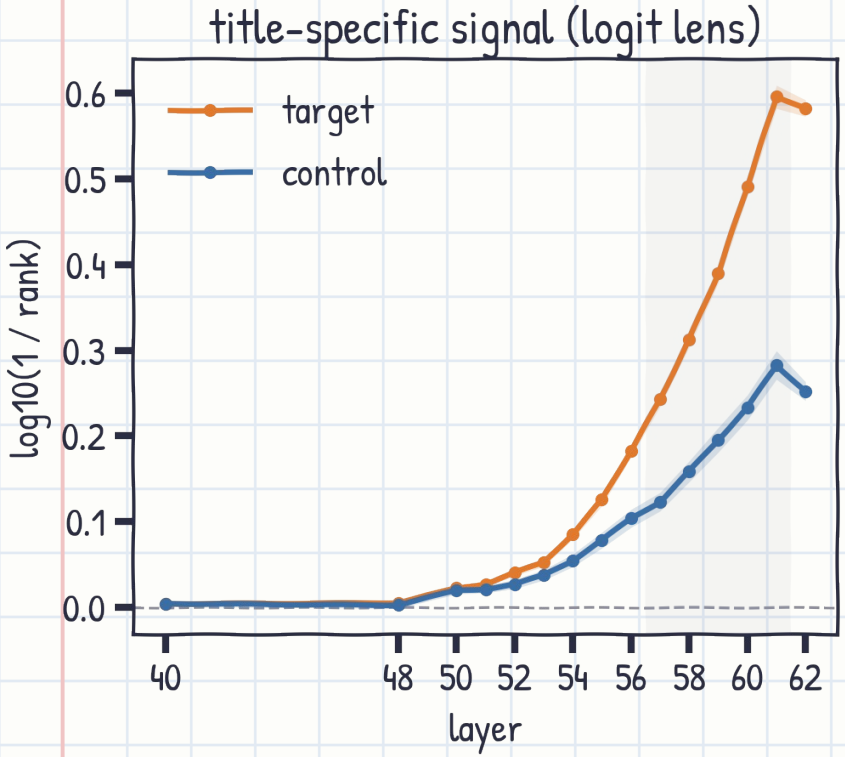
+0.21 to +0.23

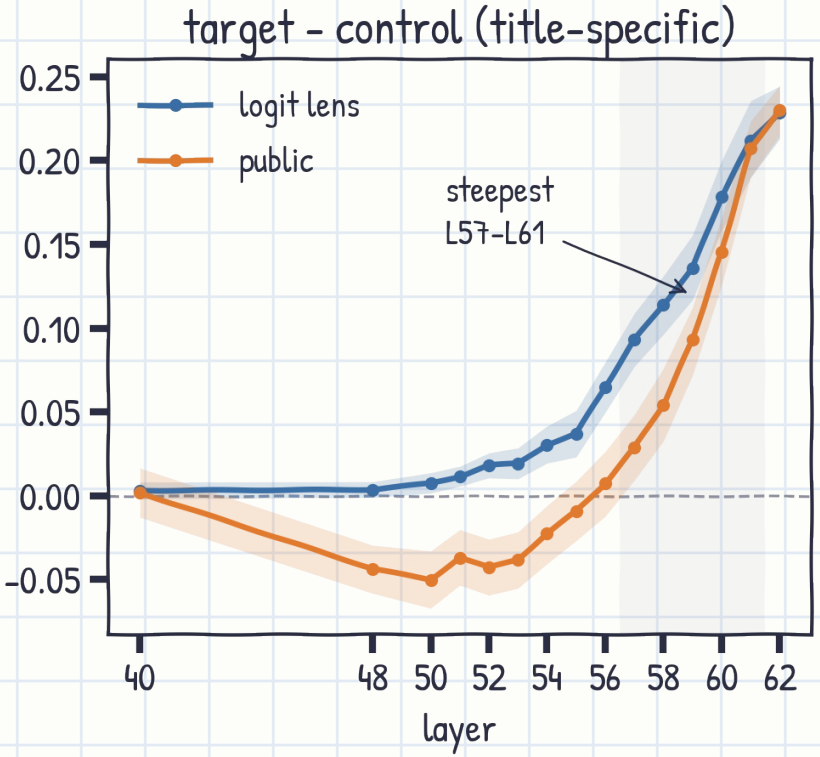
almost all title-specific

R0 only (no author written): output +0.199 vs +0.207

It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI





Logit lens: target - control

+0.007 at L50

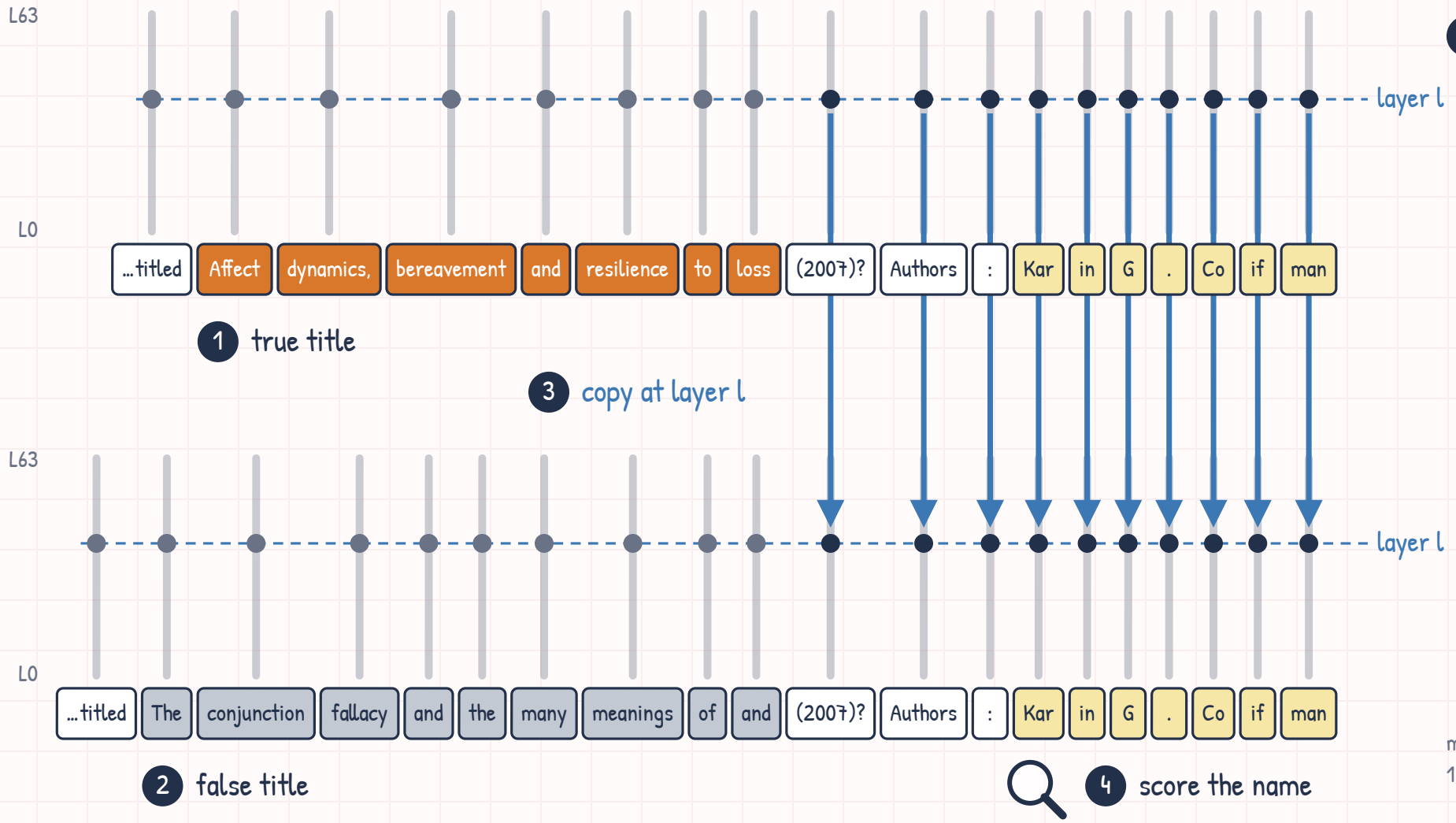
+0.23 at L62

steepest rise **L57–L61**

Replacement (patching)

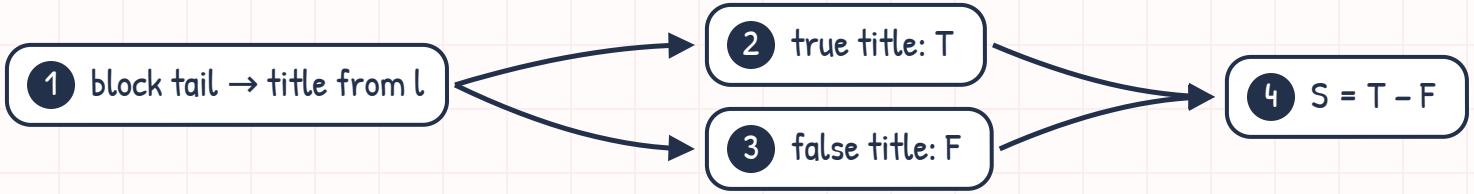
Coifman, Bonanno, Rafaeli (2007)

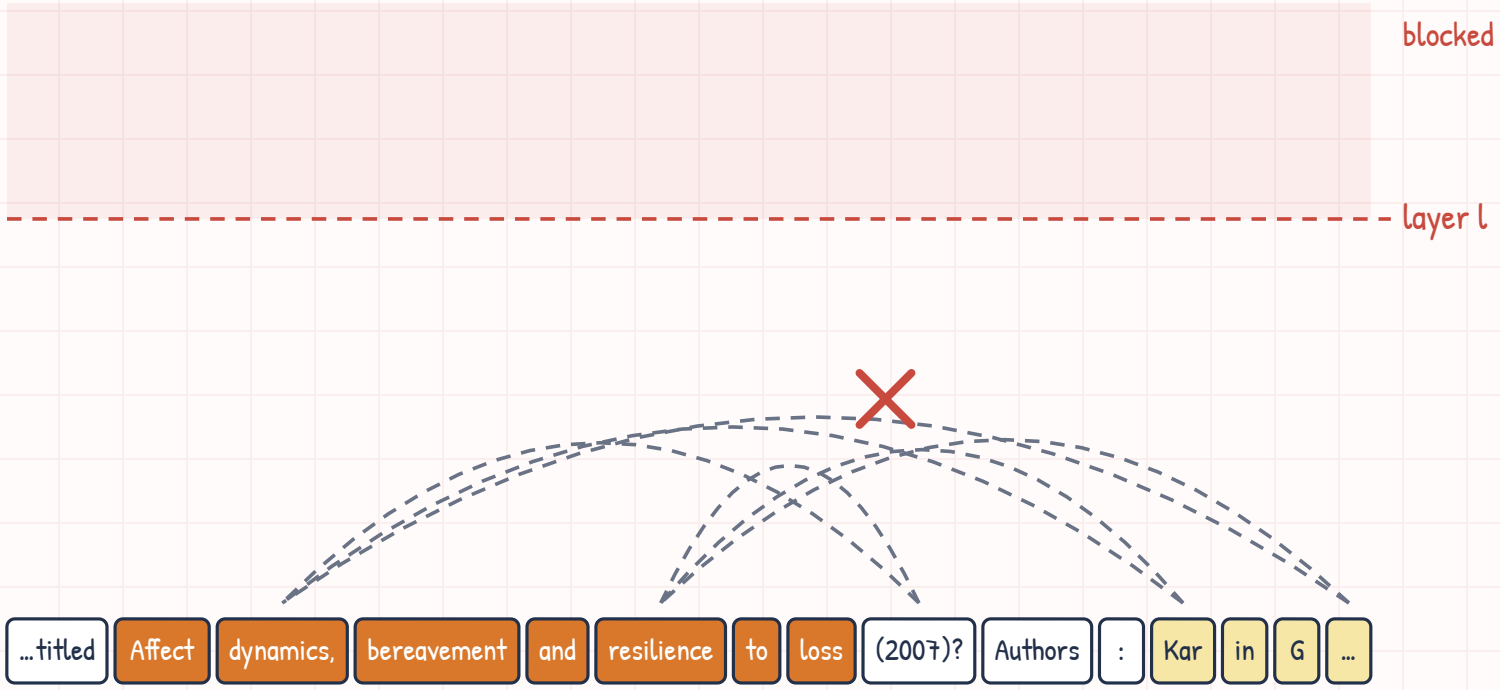
$$R(l) = (\text{patched} - \text{false}) \div (\text{true} - \text{false})$$

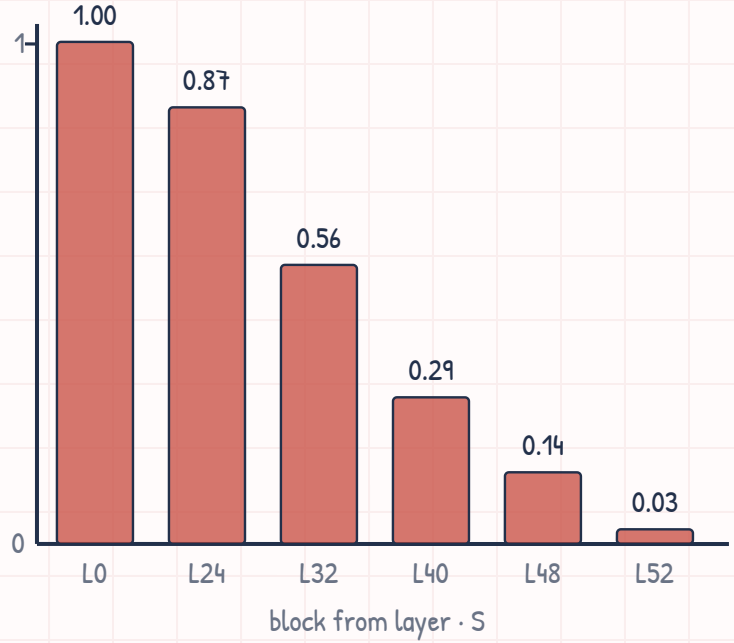


Knockout (block attention)

Coifman, Bonanno, Rafaeli (2007)







L52: target 0.018 · control 0.051

S(l) = T(l) − F(l)

loss as share of (true − false), 1,186 papers

Stage 11: intervene, do not just read

593 + 593 papers: min(60, n_t, n_c) per citation bin, seed 11 · patch uses false_0 · knockout masks all heads · all 9,108 for probe and chain

patch

copy the true-title hidden state into the false-title run at one layer

knock out

block attention to the title tokens (all heads) from layer l to the last layer

probe

linear map from the last title token to the output state

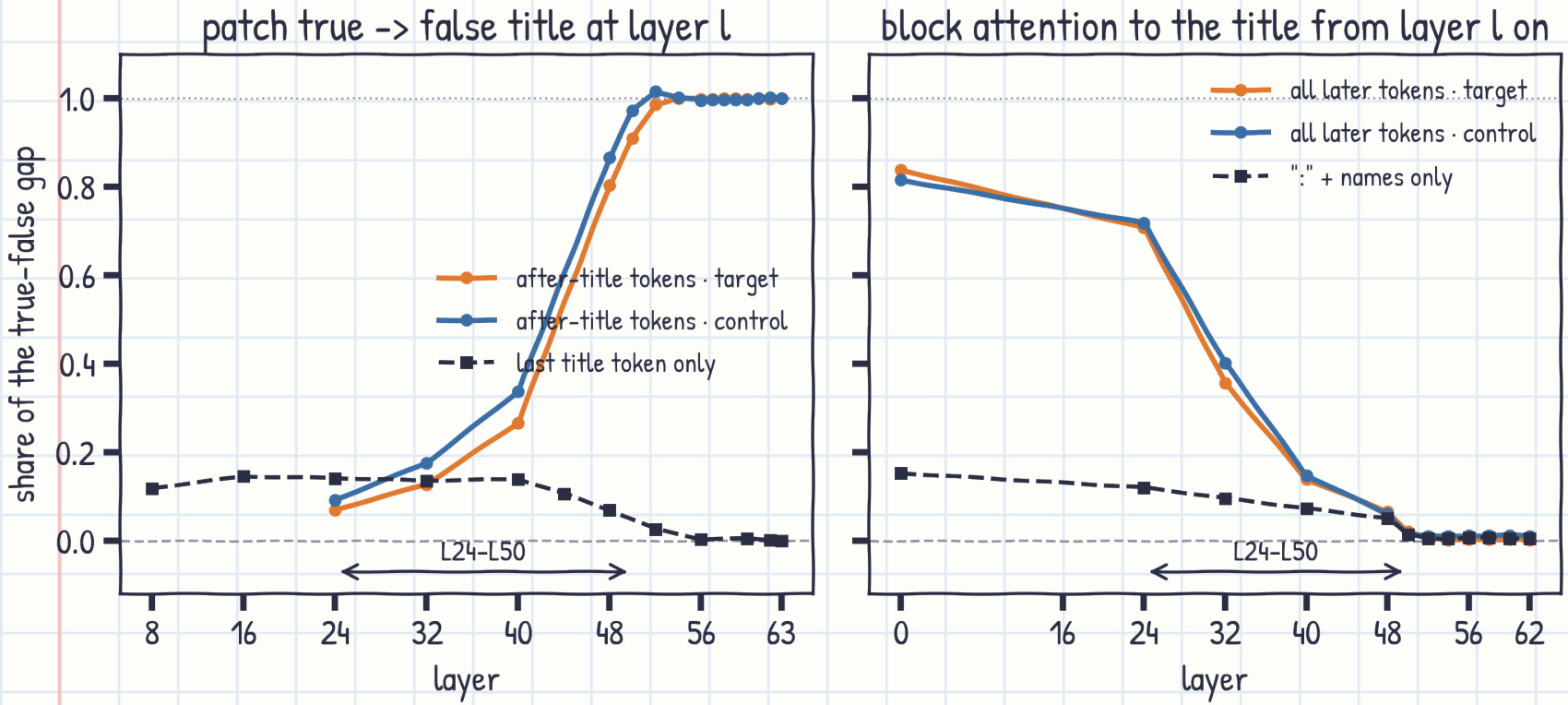
chain

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened − false) / (true − false).

The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02



L52+: title no longer needed

the information enters the **name positions at L44-52**, partly via the tokens right after the title

same timing for target and control

Stages 1–9 · 20–200 papers

What we saw

- Stage 4 peak at L56 driven by one paper (71%)
- 7.2: A–B win-rate positive at L40–L56
- 7.3, 8: L56 title advantage; L56 became the primary layer

Stages 10–11 · 9,108 papers

What scale showed

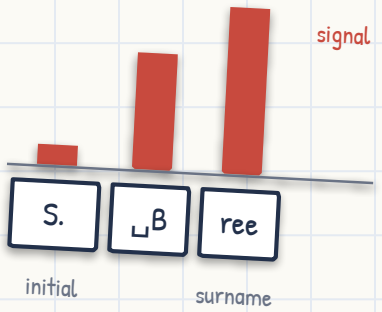
- L56 is where it *starts*; steepest rise L57–L61
- Patching: copying tail states recovers 80–87% at L48, 100% from L52
- Knockout: blocking the title from L52 on costs nothing

Line C: "read early, spoken late"

L56 was a real landmark, but not the peak. Interventions add what readouts cannot: the title is consumed in L24–50, and the answer only becomes readable at the name a dozen layers later.

Line D

Which token carries it?



small samples asked

First token or the rest of the name?

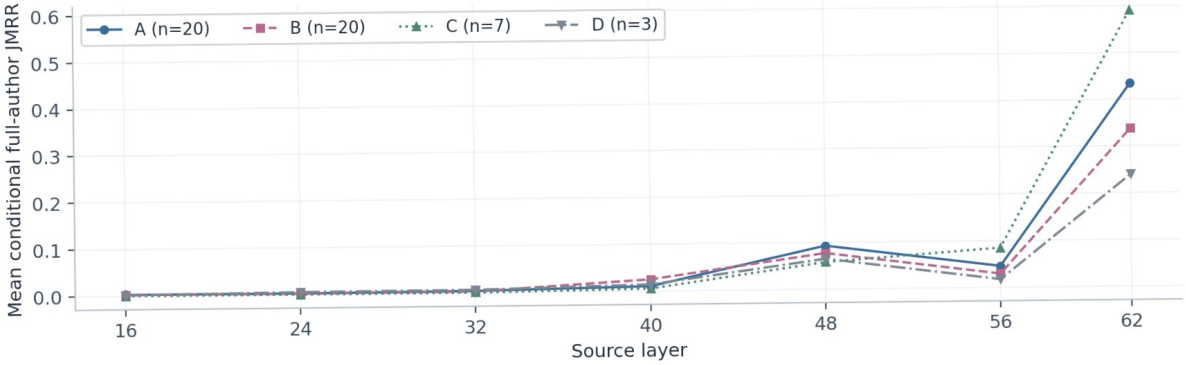
→

the full cohort answered

Depends on name format; the information enters the name positions at L44–52, partly via the tokens right after the title; co-authors partly replace the title

From first token to the full name

50 papers, 227 authors, 972 name tokens; every listed author scored with teacher forcing



Conditional full-author JMRR by group

Group	Integrated JMRR
A (20)	0.085 [0.075, 0.096]
B (20)	0.068 [0.057, 0.080]
C (7)	0.106
D (3)	0.050

L62 A–B +0.097 [0.028, 0.163]; Vanilla also +0.091

Positive, but each token after the first is given the correct prefix. Next: split first token from the rest (7.1).

The A–B gap sits on the rest of the name

CPU-only re-analysis of Stage 6; 10,000 pair-bootstrap draws



Integrated A–B JMRR by token position

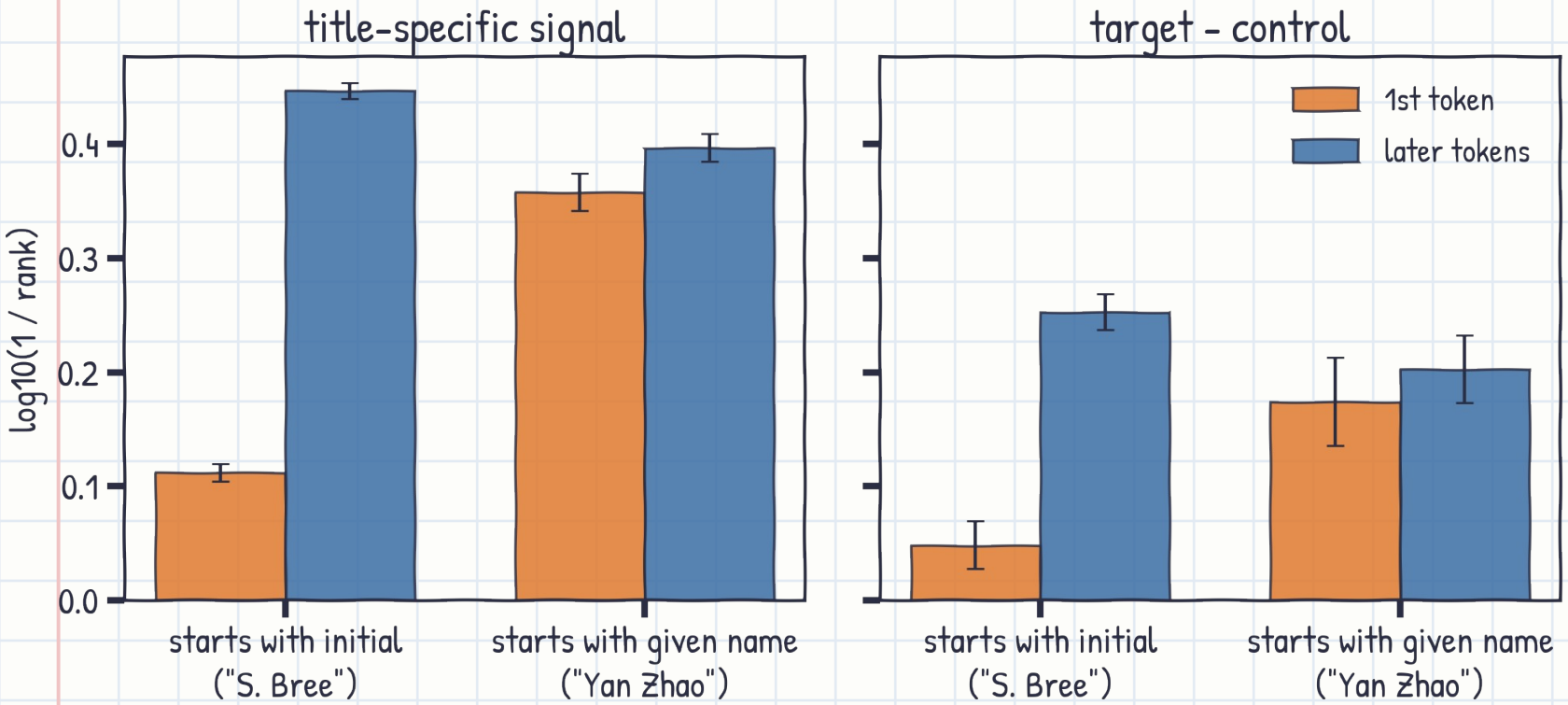
Endpoint	A–B (7 layers)
First token	+0.001 [−0.011, +0.015]
Continuation	+0.021 [+0.002, +0.039]
All tokens	+0.017 [+0.003, +0.030]

At L62: first +0.014 (12/20 pairs) vs continuation +0.121 (15/20)

Completing a name is easier than starting it. A continuation gap cannot show unaided recall. **Open:** is this about initials vs surnames?

First token or the rest of the name?

model output · split by name format

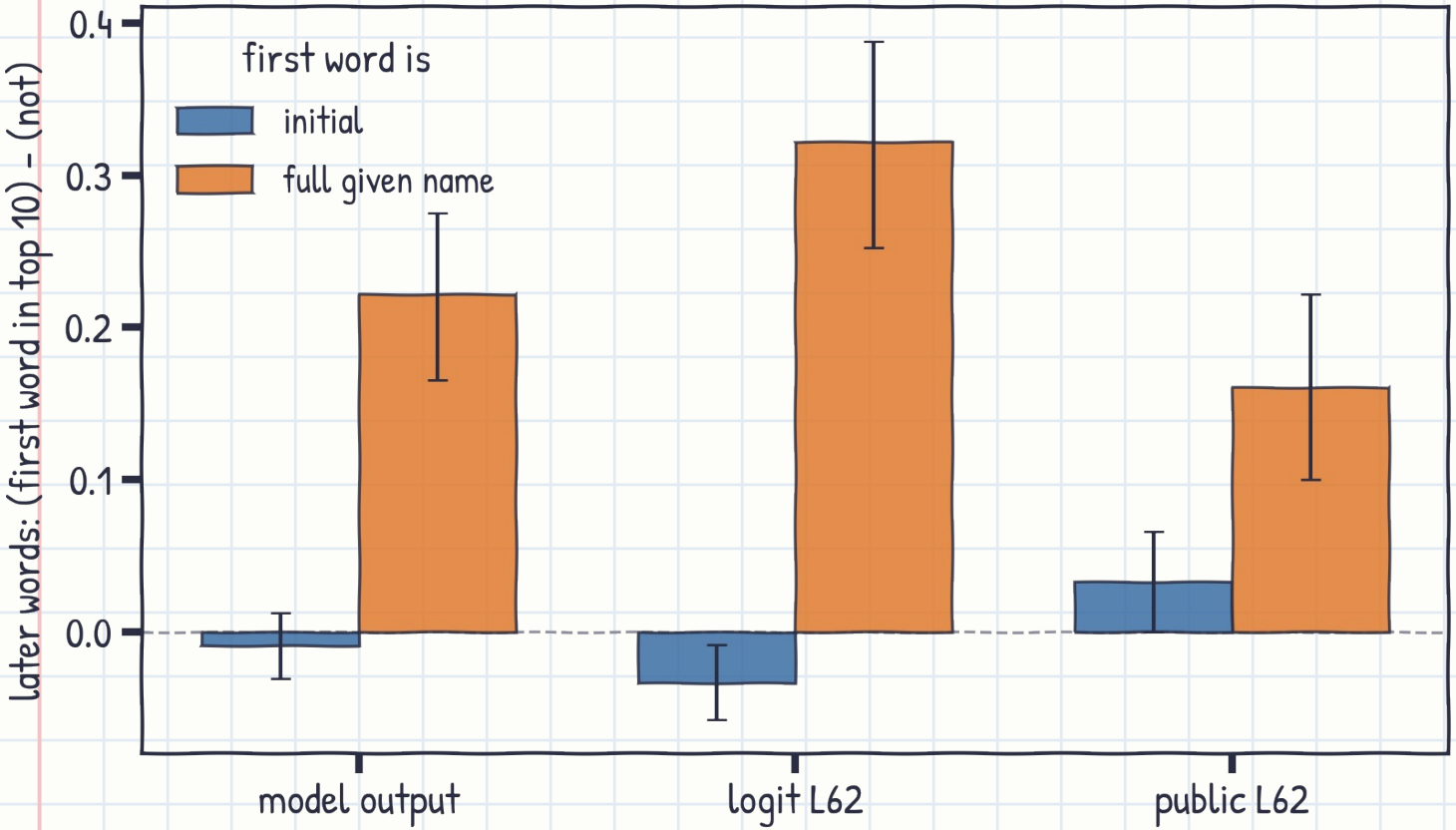


Initial: carries little; signal is on the surname

Full given name: carries about as much as the surname

Knowing the given name helps with the surname

true title · later words when the first word is / is not in the top 10



Given name in top 10

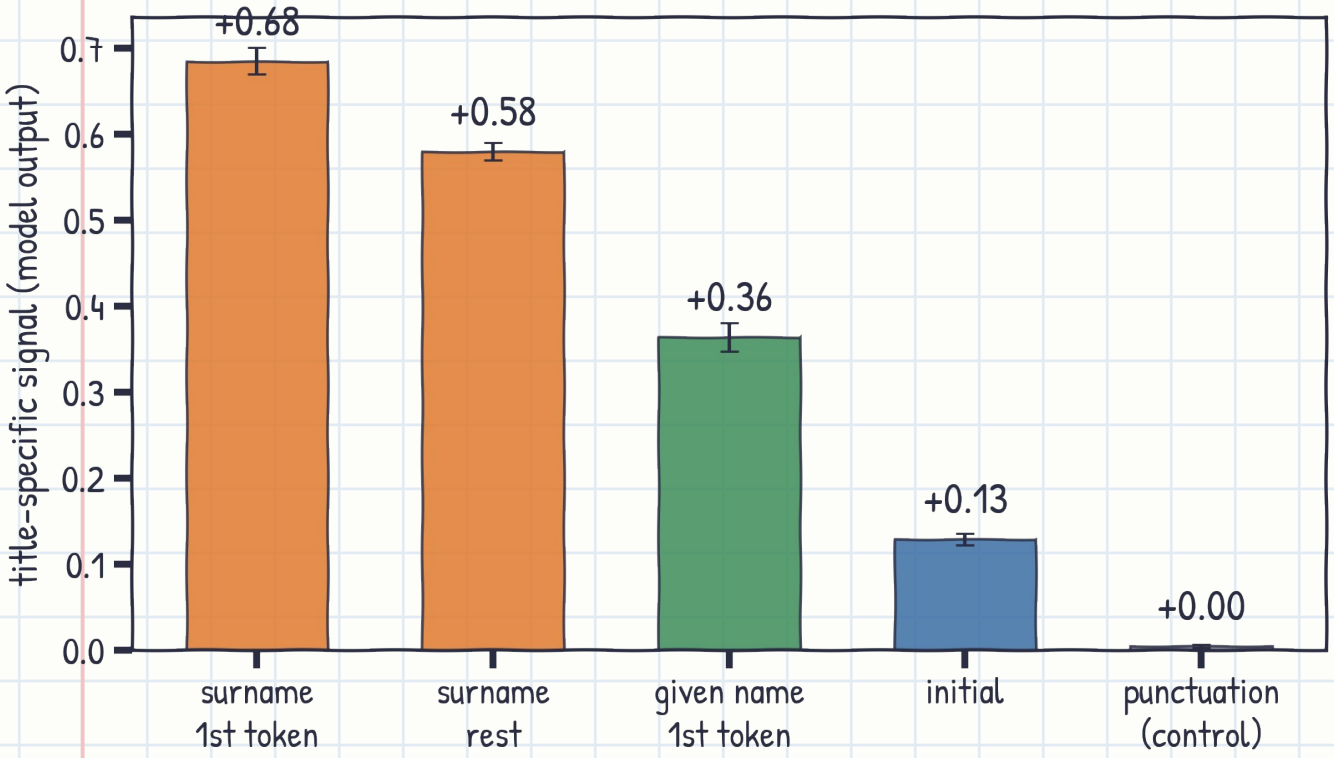
surname +0.22 (output)

+0.32 (logit L62)

Initial in top 10: no help

Where the signal sits, and a dose-response

model output



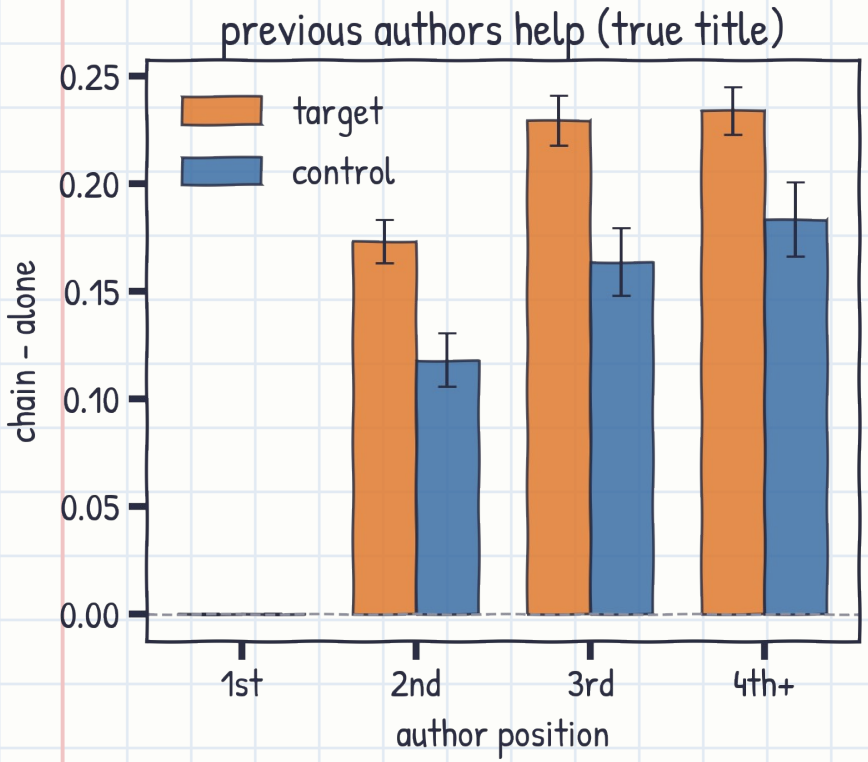


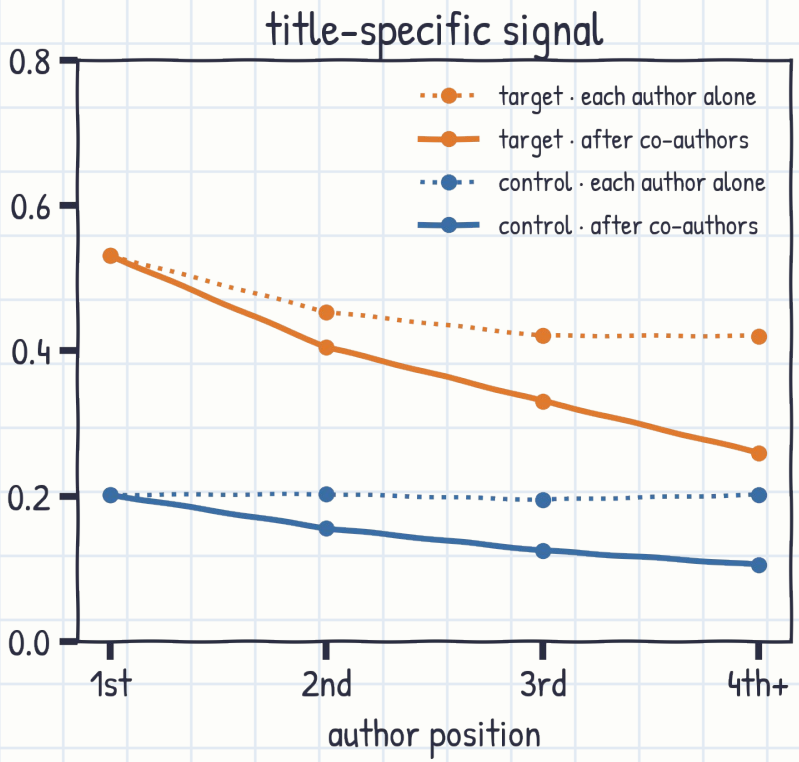
Right: R0/R1 papers, centred within citation bin; Spearman rho = 0.27, p < 0.001.

Stage 11

Co-authors help, and partly replace the title

model output · all 9,108 papers · 95% CI





later authors +0.17 to +0.23

title-specific part **shrinks**

target > control still

Stages 1–9 · 20–200 papers

What we saw

- A–B lives on continuation tokens (+0.021), not the first token (+0.001)
- Open question: is that teacher forcing, initials or surnames?

Stages 10–11 · 9,108 papers

What scale showed

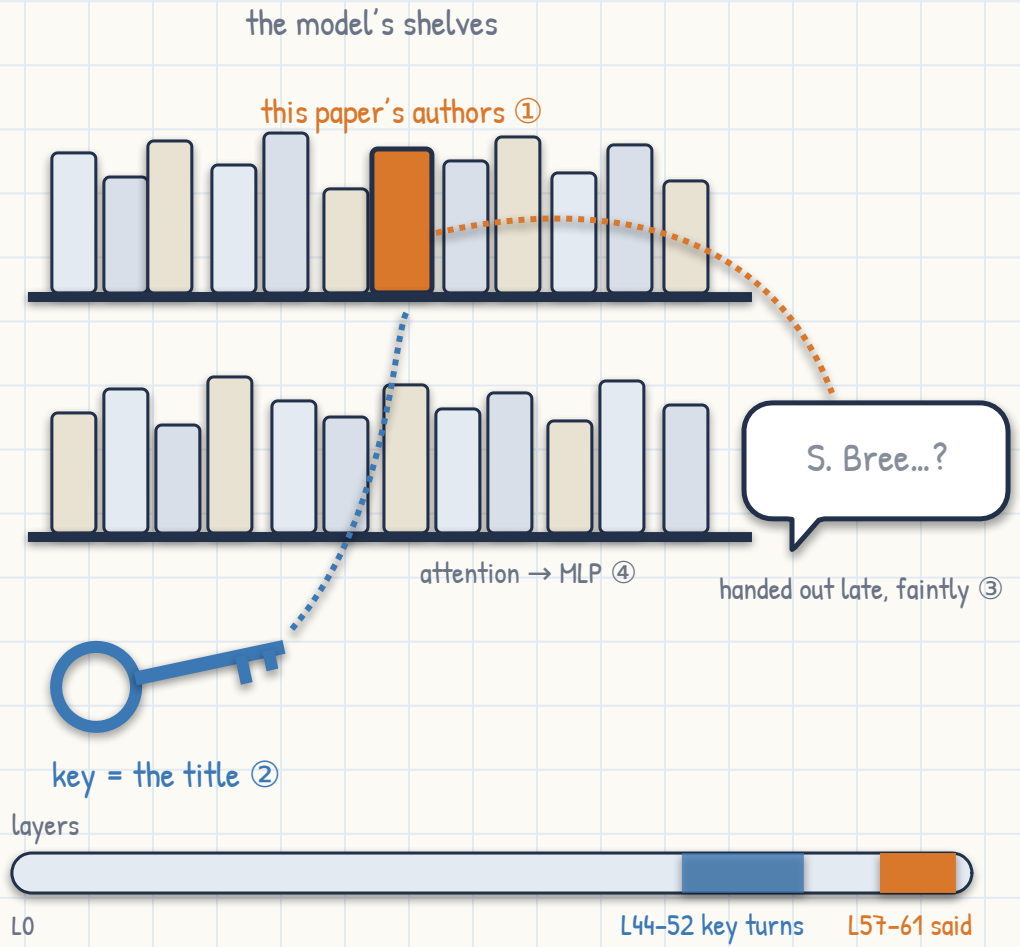
- 73% of names start with an initial, which carries little
- Full given names carry as much as surnames and help predict them (+0.22)
- Information travels title → name positions (L44–52), partly via tail tokens
- Earlier co-authors make later ones easier, title matters less

Line D: the puzzle from 7.1 is explained by name format

The pilot's "continuation" result was mostly a format effect: for "S. Bree" the surname is a continuation token. Stage 12 P1 (done) patched single positions: the name positions carry the signal (+0.85 at L52), the colon alone little (+0.17).

Keys and shelves

Is the book on the shelf, or is the key failing to get it out?



1 Is the book on the shelf? Yes

True title beats false titles (≈0.44 at L62); a probe on the title token reads paper-specific authors from L24 (re-checked)

Lines A, B · Stage 5, 7.3, 8D, 10, 11

2 Key = the title, read into the name positions at L44–52

Whole prompt: 0.83 at L48, ≈1.0 from L52; name positions alone 0.61 → 0.85; ':' alone ≤ 0.17; knockout after L52 costs almost nothing

Line C · Stage 11, 12 P1

3 Book there but stuck? Not supported

Target − control ≈ 0 in middle layers, grows only at L57–61 (+0.21 at output): retrieval itself is weak

Lines A, C · Stage 7.2, 10

4 Which shelf? Both, in order

Attention brings the title in at L48 (0.23); MLP takes over at L52–56 (0.18, 0.22). One-layer patching is a lower bound; whether the MLP step is memory lookup is not yet separable

Stage 12 P2

Caveat: a readout not finding it ≠ absent. Stage 13: ':' and name positions hold the task; the title position holds paper attributes and meaning beyond the words, not shown to be paper identity.

What the small samples taught us to watch for

- **Single-paper leverage**: one paper = 71% of the Stage 4 L56 mean
- **Tail-driven contrasts**: Stage 8 A–B disappears after removing 10 of 100 pairs
- **Forking paths**: L56 inspected repeatedly, then made primary
- **Candidate artefacts**: 100 false names share ~46 first tokens; 3% tie with gold
- **Label noise**: strict "pass" 25.5% in the 0–2 citation bin
- **Partial coverage**: first-listed author stands in for the team
- **Metric choice**: KL fidelity and author ranking disagree
- **Protocol drift**: primary endpoint changed between README and run

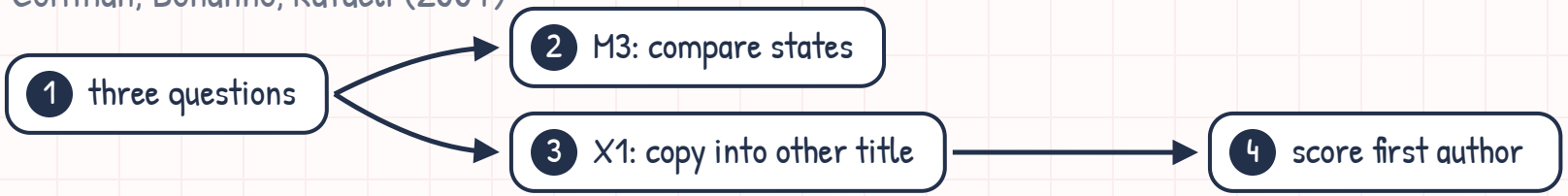
Stage 10 answered each one by design: full cohort, every author and token, all layers reported, full-vocabulary ranks, within-bin comparison, frozen inputs with parity checks. What it cannot fix: groups still come from one labelling run of the same model.

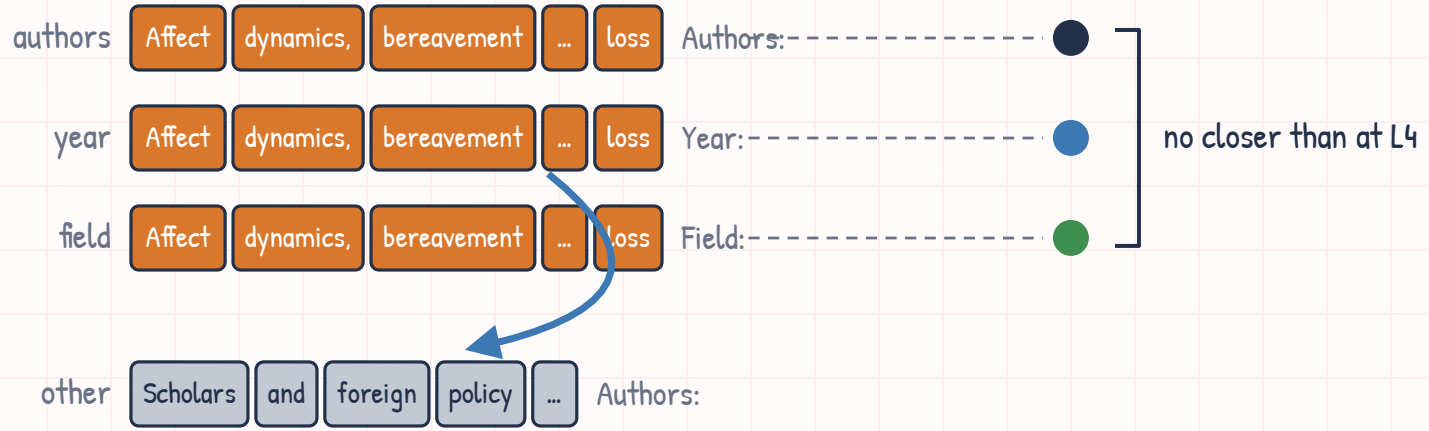
Small vs large, in one table

Finding	Stages 1–9 (20–200 papers)	Stages 10–11 (9,108 papers)	Verdict
True title helps	+9.3 pp pilot, +5.2 pp new 200	≈0.44 title-specific at L62	confirmed
Recognised > not recognised	uncertain, tail-driven	+0.21 at output, ≈0 middle	confirmed, late only
Same groups (A vs B, R0 only)	A – B as above	+0.20 at output, ≈0 middle; R1 adds strength, not timing	comparable, unchanged
Key layer	L56	starts L56, rises L57–61; read L24–50	refined
J-Lens over Vanilla	no increment	more name shape, no paper identity	consistent
First vs rest of name	gap on continuation	explained by initials vs given names	explained
Citations	low–high inconclusive	monotonic, $\rho = 0.45$	resolved
Lens stability	split-half one split; Stage 9 only Fit A	three Jacobian lenses fitted on different text agree; fit stability rests on Stage 9 Fit A only	only one piece of evidence (Stage 9 Fit A); Fit B not completed

Changing the question

Coifman, Bonanno, Rafaeli (2007)





R from	authors	year	field
L24	0.99	0.98	0.99
L40	0.92	0.92	0.92
L48	0.42	0.41	0.42

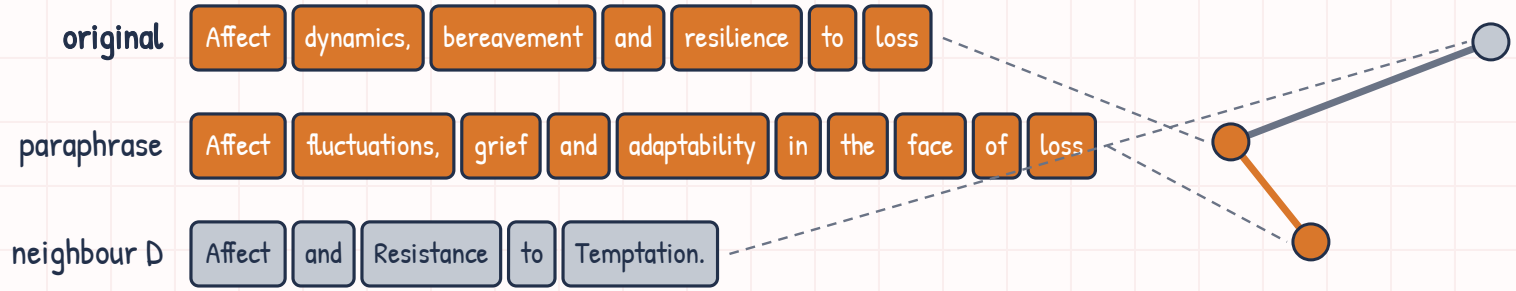
$$R(l) = (\text{patched} - \text{other}) \div (\text{own} - \text{other})$$

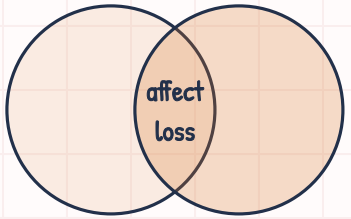
title states copied at layer l; 582 test papers

Paraphrased title (X3)

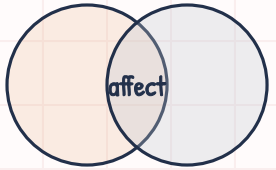
same paper in other words vs other paper, same words

1 three titles → 2 states at last title token → 3 compare distances → 4 word overlap





para 0.22



D 0.14

G″ = cos(orig, para) − cos(orig, D)

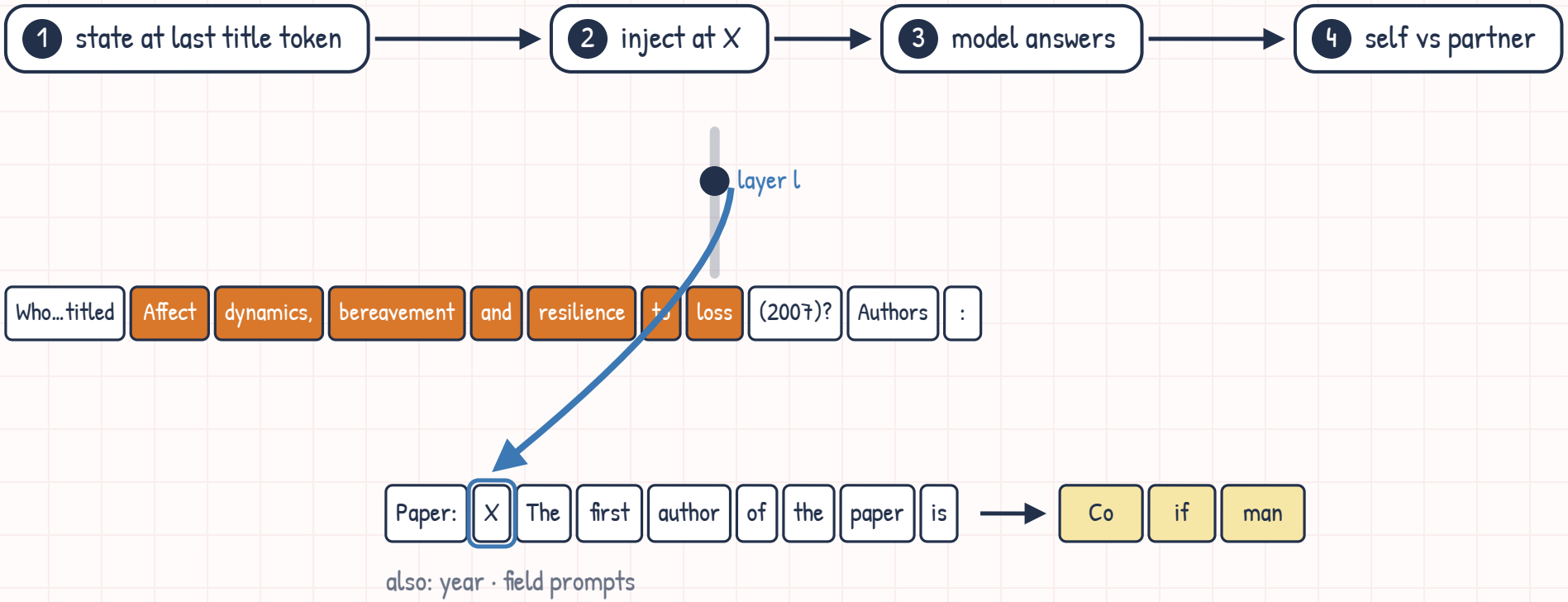
+0.32 at L8 · peak +0.46 at L36 (test set)

medians: 0.61 vs 0.15

first-author score, paraphrase − D: +0.239

Patchscope (X4)

Coifman, Bonanno, Rafaeli (2007)



D = score(own state) − score(partner state)

score = log10(1/rank) of the true answer tokens

first author from L8: +0.046

year from L40: +0.013

Stage 12–13: what the middle layers hold

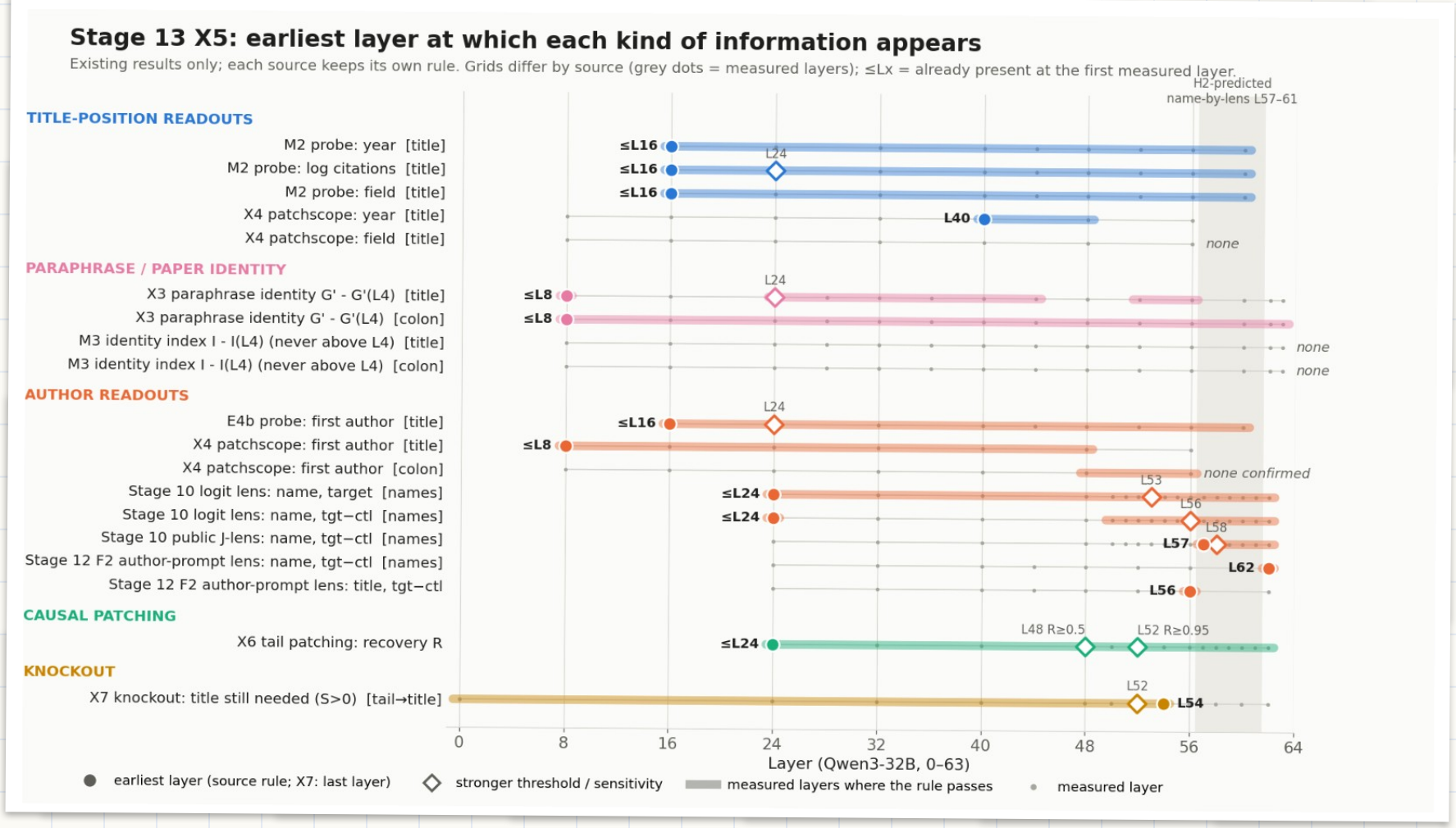
Paper facts are readable at the title early (M2) · H2 (paper identity) partly supported but not established; H3 (hidden author knowledge): insufficient evidence

假设	M1：lens 在中间层排第一的词	M2：probe 读非名字的属性	M3：换任务对照	目前结果（2026-11-01）
H1 任务 / 格式：中间层在表示“要输出作者名单”	多为 “author”、标点、名字常见开头，与标题无关	论文属性读不出或很弱	同一标题在不同问题下，中间层表示早早分开	见上文 M3、X1 结果
H2 论文身份：中间层在表示“是哪一篇论文”	与标题有关的词（照抄标题），换假标题就变	领域、年份（从标题推出）、引用数、是否认得出都读得出，且早于名字	不同问题下同一篇论文的表示在中间层相近，后期才按任务分开 **结果：**不高于 L4 文字水平，不支持	**M3 不支持；X3 显示中间层超出用词的语义表示，但不能与论文身份区分；H2 目前依据 M2 和 X1，其中 M2 的 Qwen 胜过 BERT 按补充 2 只作参考。**另外，X4 最早可解码的顺序是第一作者（L8）早于年份（L40），与 H2 预测的方向相反。
H3 作者信息已在，只是不在词表方向	不一定	作者首 token 用 probe 读得出，lens 读不出	按作者训练的 probe 能跨任务迁移；作者专用 lens（M4）在中间层读得出名字	**文本基线不弱于 Qwen，不能排除文字解释，证据不足**（X4：名字首 token 上 BERT +0.088 ⩾ Qwen L8 +0.018；L8 信号主要在姓上，见上文 X4）

Hypotheses H1–H3 × readouts M1–M3 and current results (report, 2026-11-01)

Layer order (X5)

Earliest layer at which each kind of information appears · existing results only, each source keeps its own rule



Read early, carried, spoken late

≤L8–L16: title facts and the first-author token

L48–L52: the paper signal reaches the post-title tail (patching R ≥ 0.5 at L48, ≥ 0.95 at L52; knockout costs nothing from L52)

L53–L62: names readable by lens

What we learned: read early, spoken late

Read early	Causally the title is read in L24–L50 and gathered at the name positions by L52, then no longer needed. A probe at the title token reads a little of it from L24 (re-checked).
Spoken late	At the name positions, paper-specific author knowledge becomes lens-readable only from ~L50, fastest over L57–L61.
No lens lead	Read with a lens at the names, target is not ahead of control in middle layers; the +0.21 gap is at the end (probe at the title: target − control +0.04–0.08, re-checked).
Real knowledge	Name prior is ~1/3 of the gap overall but ~0 on surnames; the rest is paper-specific and grows with citations.
Lenses agree	Jacobian lenses read more in middle layers, but nothing target-specific.
Name format	Initials carry little; full given names carry as much as surnames and help predict them.

Limits

Done: Stage 12

- Single-position patching (P1); attention vs MLP (P2, hook check passed)
- Name priors (N1, N2); author-prompt lens (F1, F2): middle layers still not readable

Done: Stage 13

- Probes read year, citations, field at the title from ≤L16 (M2); identity not above the L4 text level (M3)
- Replacement is task-independent (X1); paraphrase meaning beyond wording in L28-L40, not separable from paper identity (X3)
- First author decodable at L8, may be text-explainable (X4); 5 false titles confirm the Stage 11 timing (X6, X7)

Limits

- One model (Qwen3-32B)
- Teacher-forced preference, not generation
- Labels from the team lead's package; labelling code (elm) not available
- H2 (paper identity) partly supported but not established; H3 (hidden author knowledge): insufficient evidence